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United States Patent Application Publication

20250287650

Kind Code

A1

Publication Date

September 11, 2025

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Optimized Proximity Profile for Strained Source/Drain Feature and Method of Fabricating Thereof

Abstract

Source and drain formation techniques disclosed herein provide FinFETs with reduced channel resistance and reduced drain-induced barrier lowering. An exemplary three-step etch method for forming a source/drain recess in a source/drain region of a fin includes a first anisotropic etch, an isotropic etch, and a second anisotropic etch. The first anisotropic etch and the isotropic etch are tuned to define a location of a source/drain tip. A depth of the source/drain recess after the first anisotropic etch and the isotropic etch is less than a target depth. The second anisotropic etch is tuned to extend the depth of the source/drain recess to the target depth. The source/drain tip is near a top of the fin to reduce channel resistance while a bottom portion of the source/drain recess is spaced a distance from a gate footing that can minimize DIBL. The source/drain recess is filled with an epitaxial semiconductor material.

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Family ID: 1000008620812

Appl. No.: 19/217239

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 18446998 20230809 parent-grant-document US 12317548 child US 19217239

parent US continuation 17978576 20221101 parent-grant-document US 11824102 child US 18446998

parent US continuation 16867949 20200506 parent-grant-document US 11489062 child US

Publication Classification

Int. Cl.: **H10D62/00** (20250101); **H01L21/3065** (20060101); **H10D30/01** (20250101);
H10D30/62 (20250101); **H10D30/69** (20250101); **H10D62/13** (20250101)

U.S. Cl.:

CPC **H10D62/021** (20250101); **H01L21/3065** (20130101); **H10D30/024** (20250101);
H10D30/6211 (20250101); **H10D30/6219** (20250101); **H10D30/797** (20250101);
H10D62/151 (20250101);

Background/Summary

[0001] This application is a continuation application of U.S. patent application Ser. No. 18/446,998, filed Aug. 9, 2023, which is a continuation application of U.S. patent application Ser. No. 17/978,576, filed Nov. 1, 2022, which is a continuation application of U.S. patent application Ser. No. 16/867,949, filed May 6, 2020, which is a non-provisional application of and claims benefit of U.S. Provisional Patent Application Ser. No. 62/855,079, filed May 31, 2019, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

[0002] The integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs, where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs.

[0003] Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advances to be realized, similar developments in IC processing and manufacturing are needed. For example, as fin-like field effect transistor (FinFET) technologies progress towards smaller feature sizes (such as 32 nanometers, 28 nanometers, 20 nanometers, and below), advanced techniques are needed for precisely controlling profiles and/or dimensions of source/drain features to ensure and optimize FinFET device reliability. Although existing FinFET source/drain formation techniques have been generally adequate for their intended purposes, they have not been entirely satisfactory in all respects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 is a flow chart of a method for fabricating a fin-like field effect transistor (FinFET)

device according to various aspects of the present disclosure.

[0006] FIG. 2 is a perspective view of a FinFET device having source/drain regions, in portion or entirety, according to various aspects of the present disclosure.

[0007] FIGS. 3A-3I are fragmentary cross-sectional views of the FinFET device of FIG. 2 in a Y-Z plane along lines 1-1' at various source/drain feature fabrication stages, such as those associated with the method of FIG. 1, according to various aspects of the present disclosure.

[0008] FIGS. 4A-4I are fragmentary cross-sectional views of the FinFET device of FIG. 2 in an X-Z plane along lines 2-2' at various source/drain feature fabrication stages, such as those associated with the method of FIG. 1, according to various aspects of the present disclosure.

[0009] FIG. 5 illustrates how various source/drain feature fabrication stages can be tuned, such as those associated with the method of FIG. 1, to achieve desired profiles for source/drain features according to various aspects of the present disclosure.

[0010] FIG. 6 is a graph that illustrates channel resistance characteristics and drain-induced barrier lowering characteristics of FinFET devices having epitaxial source/drain features with different profiles according to various aspects of the present disclosure.

[0011] FIG. 7 is a fragmentary cross-sectional view of the FinFET device of FIG. 2 in a Y-Z plane along lines 1-1' at a source/drain feature fabrication stage similar to that of FIG. 3I and FIG. 4I according to various aspects of the present disclosure.

DETAILED DESCRIPTION

[0012] The present disclosure relates generally to integrated circuit devices, and more particularly, to fin-like field effect transistors (FinFETs).

[0013] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, spatially relative terms, for example, "lower," "upper," "horizontal," "vertical," "above," "over," "below," "beneath," "up," "down," "top," "bottom," etc. as well as derivatives thereof (e.g., "horizontally," "downwardly," "upwardly," etc.) are used for ease of the present disclosure of one features relationship to another feature. The spatially relative terms are intended to cover different orientations of the device including the features. Furthermore, when a number or a range of numbers is described with "about," "approximate," and the like, the term is intended to encompass numbers that are within a reasonable range considering variations that inherently arise during manufacturing. For example, the number or range of numbers encompasses numbers within a reasonable range including the number described, such as within $\pm 10\%$ of the number described or other values as understood by person skilled in the art. For example, the term "about 5 nm" encompasses the dimension range from 4.5 nm to 5.5 nm. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0014] For advanced IC technology nodes, FinFETs (also referred to as non-planar transistors) have become a popular and promising candidate for high performance and low leakage applications. As FinFETs are scaled down through various technology nodes, strained source/drain features (also referred to as stressor regions or raised source/drain features) have been formed from epitaxial semiconductor materials to enhance carrier mobility and improve device performance. Forming strained source/drain features typically includes etching recesses in source/drain regions of a fin and filling the recesses with epitaxial semiconductor material. For example, silicon may be

epitaxially grown in the source/drain recesses to form strained source/drain features for an n-type FinFET and silicon germanium may be epitaxially grown in the source/drain recesses to form strained source/drain features for a p-type FinFET.

[0015] Various techniques directed at shapes, configurations, and materials of strained source/drain features can be implemented to further improve FinFET performance. For example, a proximity between a strained source/drain feature and a channel region of a fin (defined, for example, in the fin under a gate stack) has been observed to impact FinFET operation. Conventional methods for fabricating strained source/drain features thus tend to focus on configuring a tip of the strained source/drain feature as close as possible to the channel region of the fin to reduce resistance between the strained source/drain feature and the channel region and increase mobility of carriers within the channel region, thereby improving FinFET operation. However, it has been observed that profiles of strained source/drain features fabricated by these conventional methods often increase drain-induced barrier lowering (DIBL), impeding short channel control. In some instances, the worsening DIBL has been observed to negate any operation improvements achieved by the reduced resistance.

[0016] To overcome such challenges, the present disclosure recognizes that a depth of the tip of the strained source/drain feature (in other words, a location of the tip relative to a top surface of the fin) and a proximity of a bottom portion of the strained source/drain feature also impact FinFET operation. For example, the present disclosure proposes methods for fabricating strained source/drain features that control a location of a tip of a strained source/drain feature, such that the tip is in a top portion of a fin (for example, a distance between a top surface of the fin and the tip is equal to about 10% of a fin height to about 50% of a fin height). The proposed methods further increase a distance between a gate stack and a bottom portion of the strained source/drain feature, compared to a conventional strained source/drain feature, such that a capacitance is reduced between the gate stack and the strained source/drain feature. In some implementations, the proposed strained source/drain features have a candle light bulb-shaped cross-section. The proposed strained source/drain features are achieved by performing a three-step etching process, which includes a first anisotropic etch, an isotropic etch, and a second anisotropic etch. Such three-step etching process allows for precise control of a location of the tip of the strained source/drain feature (in particular, a depth of the tip in the fin and a distance between the tip and the channel region of the fin) and a distance between a bottom portion of the strained source/drain feature and the channel region and/or the gate stack. The disclosed profile of the strained source/drain feature can reduce resistance and reduce DIBL, thereby providing a FinFET with improved operation and short channel control. Details of the proposed methods for fabricating strained source/drain features and the proposed profiles for strained source/drain features are described herein in the following description.

[0017] FIG. 1 is a flow chart of a method **100** for fabricating an integrated circuit device according to various aspects of the present disclosure. In the present embodiment, method **100** fabricates an integrated circuit device that includes a FinFET device. At block **110** and block **115**, a fin is formed over a substrate and a gate stack is formed over a channel region of the fin, respectively. The channel region of the fin is disposed between a source region of the fin and a drain region of the fin (collectively referred to as source/drain regions). At block **120**, gate spacers are formed along sidewalls of the gate stack. Block **120** can include block **122**, block **124**, and block **126**. For example, forming gate spacers can include forming a spacer layer over the gate stack and the fin at block **122**, performing a first spacer etch at block **124** (for example, a main spacer etch), and performing a second spacer etch at block **126** (for example, a spacer over etch). In some embodiments, fin spacers are formed along sidewalls of the source/drain regions of the fin. The fin spacers may be formed simultaneously with the gate spacers. At block **130**, method **100** proceeds with forming source/drain recesses in source/drain regions of the fin. Profiles of the source/drain recesses optimize FinFET performance, as described in detail herein. Block **130** includes block

132, block **134**, and block **136**. For example, forming the source/drain recesses includes performing a first anisotropic etch process at block **132**, performing an isotropic etch process at block **134**, and performing a second anisotropic etch process at block **136**. A depth of the source/drain recesses after the first anisotropic etch process and a depth of the source/drain recesses after the isotropic etch process are less than a target depth for the source/drain recesses. Parameters of the first anisotropic etch process and the isotropic etch process are tuned to define a location of tips of the source/drain recesses, such as tip depth and tip proximity, that minimize parasitic resistance between source/drain features and a metal gate of the FinFET device. Parameters of the second anisotropic etch process are tuned to extend the source/drain recesses to the target depth. Defining the location of the source/drain tips before extending the source/drain recesses to the target depth results in a bottom portion of the source/drain recesses pulling back from the metal gate during the second anisotropic etch process, increasing distance between any gate footing and the source/drain recesses, which improves short channel control. In some embodiments, formation of the source/drain recesses may begin with formation of gate spacers at block **130**, as described below.

[0018] The method **100** then proceeds to block **140** where the source/drain recesses are filled with a semiconductor material, thereby forming semiconductor source/drain features. At block **145**, a dielectric layer is formed over the fin, the gate stack, the gate spacers, and the semiconductor source/drain features. At block **150**, where method **100** includes a gate last process, a gate replacement process is performed to replace the gate stack (for example, a polysilicon dummy gate) with a metal gate stack, which can include a gate electrode disposed over a gate dielectric (e.g., a high-k dielectric layer). Alternatively, in some embodiments, method **100** includes a gate first process, where the gate stack formed at block **115** is a metal gate stack, such that method **100** may omit block **150**. In yet another alternative, method **100** includes a hybrid gate first, gate last process, which may involve forming portions of the metal gate stack at block **115**, block **150**, and/or other stage of method **100**. At block **155**, method **100** proceeds with forming contacts to the semiconductor source/drain features and/or the metal gate stack. Processing may proceed with completing fabrication of the FinFET device. Additional steps can be provided before, during, and after method **100**, and some of the steps described can be moved, replaced, or eliminated for additional embodiments of method **100**.

[0019] FIG. **2** is a perspective view of a FinFET device **200**, in portion or entirety, according to various aspects of the present disclosure. FIGS. **3A-3I** are fragmentary cross-sectional views of FinFET device **200** in a Y-Z plane along lines **1-1'** at various fabrication stages (such as those associated with method **100**) according to various aspects of the present disclosure. FIGS. **4A-4I** are fragmentary cross-sectional views of FinFET device **200** in an X-Z plane along lines **2-1'** at various fabrication stages (such as those associated with method **100**) according to various aspects of the present disclosure. FinFET device **200** may be included in a microprocessor, a memory, and/or other IC device. In some embodiments, FinFET device **200** is a portion of an IC chip, a system on chip (SoC), or portion thereof, that includes various passive and active microelectronic devices such as resistors, capacitors, inductors, diodes, p-type field effect transistors (PFETs), n-type field effect transistors (NFETs), metal-oxide semiconductor field effect transistors (MOSFETs), complementary metal-oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJTs), laterally diffused MOS (LDMOS) transistors, high voltage transistors, high frequency transistors, other suitable components, or combinations thereof. FIG. **2**, FIGS. **3A-3I**, and FIGS. **4A-4I** have been simplified for the sake of clarity to better understand the inventive concepts of the present disclosure. Additional features can be added in FinFET device **200**, and some of the features described below can be replaced, modified, or eliminated in other embodiments of FinFET device **200**.

[0020] In FIG. **2**, FIG. **3A**, and FIG. **4A**, FinFET device **200** includes a substrate (wafer) **210**. In the depicted embodiment, substrate **210** includes silicon. Alternatively or additionally, substrate **210** includes another elementary semiconductor, such as germanium; a compound semiconductor,

such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium antimonide, and/or indium antimonide; an alloy semiconductor, such as silicon germanium (SiGe), GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Alternatively, substrate **210** is a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator (SGOI) substrate, or a germanium-on-insulator (GOI) substrate. Semiconductor-on-insulator substrates can be fabricated using separation by implantation of oxygen (SIMOX), wafer bonding, and/or other suitable methods. Substrate **210** can include various doped regions depending on design requirements of FinFET device **200**. For example, substrate **210** can include a p-type doped region (referred to as a p-well), which can be configured for an n-type FinFET, and an n-type doped region (referred to as an n-well), which can be configured for a p-type FinFET. N-type doped regions are doped with n-type dopants, such as phosphorus, arsenic, other n-type dopant, or combinations thereof. P-type doped regions are doped with p-type dopants, such as boron, indium, other p-type dopant, or combinations thereof. In some embodiments, substrate **210** includes doped regions formed with a combination of p-type dopants and n-type dopants. The various doped regions can be formed directly on and/or in substrate **210**, for example, providing a p-well structure, an n-well structure, a dual-well structure, a raised structure, or combinations thereof. An ion implantation process, a diffusion process, and/or other suitable doping process can be performed to form the various doped regions of FinFET device **200**. [0021] A fin structure **220** is formed over substrate **210**. In FIG. 2, fin structure **220** includes multiple fins, such as a fin **222A** and a fin **222B**, extending from substrate **210**. The present disclosure contemplates embodiments where fin structure **220** includes more fins or a single fin extending from substrate **210**. Fins **222A**, **222B** extend substantially parallel to one another along a y-direction and having a length defined in the y-direction, a width defined in an x-direction, and a height defined in the z-direction. For example, a fin height FH of fins **222A**, **222B** is defined between a top surface of isolation feature **225** and respective top surfaces of fins **222A**, **222B** along the z-direction. In some embodiments, fin height FH is about 40 nm to about 70 nm. Fins **222A**, **222B** each have a channel region (C), a source region (S/D), and a drain region (S/D) defined along their length (here, along the y-direction), where the channel region is disposed between the source region and the drain region, both of which are collectively referred to as source/drain (S/D) regions. In some embodiments, fins **222A**, **222B** are a portion of substrate **210**. For example, in the depicted embodiment, where substrate **210** includes silicon, fins **222A**, **222B** include silicon. Alternatively, fins **222A**, **222B** are defined in a material layer, such as a semiconductor material layer, disposed on substrate **210**. The semiconductor material may be silicon, germanium, silicon germanium, III-V semiconductor material, other suitable semiconductor material, or combinations thereof. In some embodiments, fins **222A**, **222B** include a stack of semiconductor layers disposed over substrate **210**. The semiconductor layers can include same or different materials, dopants, etching rates, constituent atomic percentages, constituent weight percentages, thicknesses, and/or configurations depending on design requirements of FinFET device **200**.

[0022] A combination of deposition, lithography, and/or etching processes are performed to define fins **222A**, **222B** extending from substrate **210**. For example, forming fins **222A**, **222B** includes performing a lithography process to form a patterned mask layer over substrate **210** (or a material layer disposed over substrate **210**) and performing an etching process to transfer a pattern defined in the patterned mask layer to substrate **210** (or the material layer). The lithography process can include forming a resist layer over substrate **210** (for example, by spin coating), performing a pre-exposure baking process, performing an exposure process using a mask, performing a post-exposure baking process, and performing a developing process. During the exposure process, the resist layer is exposed to radiation energy (such as ultraviolet (UV) light, deep UV (DUV) light, or extreme UV (EUV) light), where the mask blocks, transmits, and/or reflects radiation to the resist layer depending on a mask pattern of the mask and/or mask type (for example, binary mask, phase shift mask, or EUV mask), such that an image is projected onto the resist layer that corresponds

with the mask pattern. Since the resist layer is sensitive to radiation energy, exposed portions of the resist layer chemically change, and exposed (or non-exposed) portions of the resist layer are dissolved during the developing process depending on characteristics of the resist layer and characteristics of a developing solution used in the developing process. After development, the patterned resist layer includes a resist pattern that corresponds with the mask. In some embodiments, the patterned resist layer is the patterned mask layer. In such embodiments, the patterned resist layer is used as an etch mask to remove portions of substrate **210** (or the material layer). In some embodiments, the patterned resist layer is formed over a mask layer formed over substrate **210** before forming the resist layer, and the patterned resist layer is used as an etch mask to remove portions of the mask layer formed over substrate **210**. In such embodiments, the patterned mask layer is used as an etch mask to remove portions of substrate **210** (or the material layer). The etching process can include a dry etching process, a wet etching process, other suitable etching process, or combinations thereof. In some embodiments, a reactive ion etching (RIE) process is used to form fins **222A**, **222B**. After the etching process, the patterned resist layer is removed from substrate **210**, for example, by a resist stripping process. In some embodiments, after the etching process, patterned mask layer is removed from substrate **210** (in some embodiments, by a resist stripping process). In some embodiments, the patterned mask layer is removed during etching of substrate **210** (or the material layer). Alternatively, fins **222A**, **222B** are formed by a multiple patterning process, such as a double patterning lithography (DPL) process (for example, a lithography-etch-lithography-etch (LELE) process, a self-aligned double patterning (SADP) process, a spacer-is-dielectric (SID) SADP process, other double patterning process, or combinations thereof), a triple patterning process (for example, a lithography-etch-lithography-etch-lithography-etch (LELELE) process, a self-aligned triple patterning (SATP) process, other triple patterning process, or combinations thereof), other multiple patterning process (for example, self-aligned quadruple patterning (SAQP) process), or combinations thereof. In some embodiments, directed self-assembly (DSA) techniques are implemented for forming fins **222A**, **222B**. Further, in some alternate embodiments, the exposure process can implement maskless lithography, e-beam writing, and/or ion-beam writing for patterning.

[0023] An isolation feature(s) **225** is formed over and/or in substrate **210** to separate and isolate various regions of FinFET device **200**, such as fin **222A** from fin **222B**. In the depicted embodiment, isolation feature **225** surrounds a lower portion L of fins **222A**, **222B**. Upper portions U of fins **222A**, **222B** extend from isolation feature **225** and substrate **210** along the z-direction, such that top surfaces of fins **222A**, fins **222B** are disposed above a top surface of isolation feature **225** along the z-direction. Isolation feature **225** includes silicon oxide, silicon nitride, silicon oxynitride, other suitable isolation material (for example, including silicon, oxygen, nitrogen, carbon, or other suitable isolation constituent), or combinations thereof. Isolation feature **225** can be configured as a shallow trench isolation (STI) structure, a deep trench isolation (DTI) structure, and/or local oxidation of silicon (LOCOS) structure. For example, isolation feature **225** may be an STI feature that defines and electrically isolates fins **222A**, **222B** from other device regions (such as fins). The STI feature can be formed by etching a trench in substrate **210** (for example, by using a dry etching process and/or a wet etching process) and filling the trench with insulator material (for example, using a chemical vapor deposition (CVD) process or a spin-on glass process). A chemical mechanical polishing (CMP) process may be performed to remove excess insulator material and/or planarize the top surface of isolation feature **225**. In another example, the STI feature can be formed by depositing an insulator material over substrate **210** after forming fins **222A**, **222B** (in some embodiments, such that the insulator material layer fills gaps (trenches) between fins **222A**, **222B**) and etching back the insulator material layer to form isolation feature **225**. In some embodiments, the STI feature includes a multi-layer structure that fills the trenches, such as a silicon nitride layer disposed over a thermal oxide liner layer. In another example, the STI feature includes a dielectric layer disposed over a doped liner layer (including, for example, boron

silicate glass (BSG) or phosphosilicate glass (PSG)). In yet another example, the STI feature includes a bulk dielectric layer disposed over a liner dielectric layer, where the bulk dielectric layer and the liner dielectric layer include materials depending on design requirements of FinFET device **200**.

[0024] Gate stacks, such as a gate stack **230A** and a gate stack **230B**, are formed over portions of fins **222A**, **222B** and over isolation feature **225**. Gate stacks **230A**, **230B** extend lengthwise in a direction that is different than (e.g., orthogonal to) the lengthwise direction of fins **222A**, **222B**. For example, gate stacks **230A**, **230B** extend substantially parallel to one another along the x-direction, having a length defined in the x-direction, a width defined in the y-direction, and a height defined in the z-direction. Gate stacks **230A**, **230B** are disposed between S/D regions of fins **222A**, **222B**, where channel regions of fins **222A**, **222B** underlie gate stacks **230A**, **230B**. In the X-Z plane, gate stacks **230A**, **230B** wrap top surfaces and sidewall surfaces of fins **222A**, **222B**. In the Y-Z plane, gate stacks **230A**, **230B** are disposed over top surfaces of respective channel regions of fins **222A**, **222B**. In the depicted embodiment, gate stacks **230A**, **230B** are dummy gate stacks that include a dummy gate electrode **232** and a hard mask **234** disposed over dummy gate electrode **232**. Dummy gate electrode **232** includes a suitable dummy gate material, such as a polysilicon layer. Gate stacks **230A**, **230B** can thus be referred to as poly (PO) gate stacks, in some embodiments. Hard mask layer **234** includes silicon oxide, silicon carbide, silicon nitride, other suitable hard mask material, or combinations thereof. In some embodiments, gate stacks **230A**, **230B** further includes a gate dielectric disposed between dummy gate electrode **232** and fins **222A**, **222B**, where the gate dielectric includes a dielectric material, such as silicon oxide, a high-k dielectric material, other suitable dielectric material, or combinations thereof. In some embodiments, the gate dielectric includes an interfacial layer (for example, a silicon oxide layer) disposed over fins **222A**, **222B** and a high-k dielectric layer disposed over the interfacial layer. Gate stacks **230A**, **230B** can include numerous other layers, for example, capping layers, interface layers, diffusion layers, barrier layers, or combinations thereof. Gate stacks **230A**, **230B** are formed by deposition processes, lithography processes, etching processes, other suitable processes, or combinations thereof. For example, a deposition process is performed to form a dummy gate electrode layer over fins **222A**, **222B** and isolation feature **225** and a hard mask layer over the dummy gate electrode layer. In some embodiments, before forming the dummy gate electrode layer, a deposition process is performed to form a gate dielectric layer over fins **222A**, **222B** and/or isolation feature **225**. In such embodiments, the dummy gate electrode layer is deposited over the gate dielectric layer. The deposition process includes CVD, physical vapor deposition (PVD), atomic layer deposition (ALD), high density plasma CVD (HDPCVD), metal organic CVD (MOCVD), remote plasma CVD (RPCVD), plasma enhanced CVD (PECVD), plasma enhanced ALD (PEALD), low-pressure CVD (LPCVD), atomic layer CVD (ALCVD), atmospheric pressure CVD (APCVD), other suitable methods, or combinations thereof. A gate patterning process (including, for example, various lithography processes, etching processes, and/or cleaning processes) are then performed to pattern the dummy gate electrode layer and the hard mask layer (and, in some embodiments, the gate dielectric layer) to form gate stacks **230A**, **230B** as depicted in FIG. 2.

[0025] Gate stacks **230A**, **230B** have a gate length $L_{sub.g1}$, which defines a distance (or length) that current (e.g., carriers, such as electrons or holes) travels between S/D regions of fins **222A**, **222B**. Gate stacks **230A**, **230B** have a gate footing GF (also referred to as gate skirt or gate ledge) disposed along bottoms of gate stacks **230A**, **230B**, which results in gate stacks **230A**, **230B** having a gate length $L_{sub.g2}$ that is greater than gate length $L_{sub.g1}$ along bottoms of gate stacks **230A**, **230B**. Gate footing GF has a tapered width that increases along the z-direction, such that gate length $L_{sub.g2}$ also increases along the z-direction. Gate footing GF (or a bottom gate profile) of gate stacks **230A**, **230B** may vary depending on patterning environment (e.g., isolated pattern or dense pattern), position of gate stacks **230A**, **230B** on a wafer (e.g., edge or center), and/or proximity of area of gate stacks **230A**, **230B** relative to fins **222A**, **222B** (e.g., gate area directly

adjacent to fins **222A**, **222B** or gate area at end of gate stacks **230A**, **230B**). In some embodiments, as depicted, gate footing GF extends along an entire width of gate stacks **230A**, **230B**. In some embodiments, gate footing GF extends partially along the width of gate stacks **230A**, **230B**. In some embodiments, gate footing GF is present only at an interface region IF between fins **222A**, **222B** and gate stacks **230A**, **230B**, such as at corners between fins **222A**, **222B** and gate stacks **230A**, **230B**. Gate footing GF can result unintentionally from the inability of conventional etching processes to achieve substantially vertical sidewalls for high aspect ratio gate patterns. High aspect ratio gate patterns generally refer to gate patterns having a ratio of gate height (e.g., along the z-direction) to gate length (e.g., along the y-direction) that is greater than about 5 (i.e., gate height/gate length ≥ 5). More recently, gate patterning processes have been tuned to intentionally form gate footing GF to minimize (or eliminate) gate collapse concerns that arise for high aspect ratio gate patterns. However, gate footing GF presents new challenges for source/drain fabrication as device dimensions shrink. For example, conventional source/drain fabrication techniques fail to consider proximity of source/drain features to gate footing GF, which affects capacitive coupling between the source/drain features and eventual metal gates of FinFETs. As proximity between source/drain features and gate footing GF decreases, parasitic capacitance between the source/drain features and gate footing GF increases, thereby degrading FinFET performance. In some instances, conventional source/drain fabrication techniques may expose gate footing GF, causing source/drain features to physically contact gate footing GF and result in electrical shorts. The proposed source/drain fabrication techniques thus account for proximity of the source/drain features to gate footing GF and achieve source/drain profiles that minimize parasitic capacitance between the source/drain features and gate footing GF. This has been observed to provide better control over short channel effects of a FinFET device, such as by reducing DIBL, as described further below.

[0026] In FIGS. **3B-3D** and FIGS. **4B-4D**, gate spacers are formed along sidewalls of gate stacks **230A**, **230B**. In some embodiments, fin spacers are also formed along sidewalls of fins **222A**, **222B**. Turning to FIG. **3B** and FIG. **4B**, a spacer layer **240** is formed over FinFET device **200** (for example, at block **122** of method **100**). For example, a deposition process is performed to deposit a dielectric layer over fins **222A**, **222B**, isolation feature **225**, and gate stacks **230A**, **230B**. The deposition process is CVD, PECVD, ALD, PEALD, PVD, other suitable deposition process, or combinations thereof. The dielectric material can include silicon, oxygen, carbon, nitrogen, other suitable spacer constituent, or combinations thereof (for example, silicon oxide, silicon nitride, silicon oxynitride (SiON), silicon carbide, silicon carbon nitride (SiCN), silicon oxycarbide (SiOC), silicon oxycarbon nitride (SiOCN), silicon boron carbon nitride (SiBCN), etc.). In the depicted embodiment, spacer layer **240** includes silicon and nitrogen (and thus may be referred to as a silicon nitride layer). In some embodiments, spacer layer **240** is a single layer, such as one silicon nitride layer. In some embodiments, spacer layer **240** includes multiple layers, such as a first dielectric layer disposed over a second dielectric layer. For example, the first dielectric layer can include silicon carbon nitride and the second dielectric layer can include silicon nitride. In furtherance of the depicted embodiment, a thickness t of spacer layer **240** is about 5 nm to about 15 nm. In some embodiments, spacer layer **240** has a substantially uniform thickness along top surfaces and sidewall surfaces of gate stacks **230A**, **230B** and fins **222A**, **222B**. For example, a thickness of spacer layer **240** along top surfaces of gate stacks **230A**, **230B** and top surface of isolation feature **225** is substantially the same as a thickness of spacer layer **240** along sidewall surfaces of gate stacks **230A**, **230B**. A thickness of spacer layer **240** along top surfaces of fins **222A**, **222B** and top surface of isolation layer **225** is also substantially the same as a thickness of spacer layer **240** along sidewall surfaces of fins **222A**, **222B**. In some embodiments, spacer layer **240** is formed by a conformal deposition process, such that spacer layer **240** conforms to surfaces of FinFET device **200** upon which spacer layer **240** is deposited (and may thus be referred to as a conformal spacer layer).

[0027] Processing then proceeds with performing a spacer etch process (for example, at block **124**

and block **126** of method **100**) on spacer layer **240** to form gate spacers **242** along sidewalls of gate stacks **230A**, **230B** and fin spacers **244** along sidewalls of fins **222A**, **222B**. The spacer etch process includes a first spacer etch (FIG. **3C** and FIG. **4C**, referred to as a spacer main etch (SME)) and a second spacer etch (FIG. **3D** and FIG. **4D**, referred to as a spacer over etch (SOE)). During the spacer etch process, first spacer etch and/or second spacer etch may unintentionally or intentionally remove a portion of fins **222A**, **222B** in the S/D regions, thereby beginning formation of source/drain recesses **250** in the S/D regions of fins **222A**, **222B**. In some embodiments, after forming gate spacers **242**, a depth of source/drain recesses **250** is less than about 20% of fin height FH. For example, source/drain recesses **250** have a depth d1 after first spacer etch (FIG. **3C** and FIG. **4C**) and a depth d2 after second spacer etch (FIG. **3D** and FIG. **4D**). Depth d1 is measured between respective top surfaces of fins **222A**, **222B** and respective portions of fins **222A**, **222B** defining bottommost surfaces of source/drain recesses **250** after first spacer etch. Depth d2 is measured between respective top surfaces of fins **222A**, **222B** and respective portions of fins **222A**, **222B** defining bottommost surfaces of source/drain recesses **250** after second spacer etch. In some embodiments, depth d1 is less than or equal to about 10% of fin height FH and depth d2 is about 10% to about 20% of fin height FH. In some embodiments, depth d1 is about 4 nm to about 10 nm. In some embodiments, depth d2 is about 4 nm to about 15 nm. The present disclosure also contemplates embodiments where first spacer etch and/or second spacer etch do not remove portions of fins **222A**, **222B** in the S/D regions, such that depth d1 and/or depth d2 may be equal to 0, in some embodiments.

[0028] Turning to FIG. **3C** and FIG. **4C**, first spacer etch selectively removes spacer layer **240** with respect to gate stacks **230A**, **230B**, isolation feature **225**, and/or fins **222A**, **222B**. In other words, first spacer etch substantially removes spacer layer **240** but does not remove, or does not substantially remove, gate stacks **230A**, **230B**, isolation feature **225**, and/or fins **222A**, **222B**. First spacer etch removes spacer layer **240** substantially from horizontal (lateral) surfaces of FinFET device **200**, such as top surfaces of fins **222A**, **222B**, top surfaces of isolation feature **225**, and top surfaces of gate stacks **230A**, **230B**, thereby forming gate spacers **242** having thickness t1 along sidewalls of gate stacks **230A**, **230B** and fin spacers **244** along sidewalls of fins **222A**, **222B**. In some embodiments, thickness t1 is about 3 nm to about 10 nm. First spacer etch may also remove portions of fins **222A**, **222B** in S/D regions to begin formation of source/drain recesses **250**. For example, source/drain recesses **250** have depth d1 after first spacer etch. In some embodiments, gate spacers **242** includes more than one set of spacers, such as seal spacers, offset spacers, sacrificial spacers, dummy spacers, and/or main spacers depending on design and/or processing requirements for FinFET device **200**.

[0029] First spacer etch is an anisotropic etch process, which generally refers to an etch process having different etch rates in different directions, such that the etch process removes material in specific directions, such as substantially in one direction. For example, first spacer etch has a vertical etch rate that is greater than a horizontal etch rate (in some embodiments, the horizontal etch rate equals zero). First spacer etch thus removes material in substantially the vertical direction (here, z-direction) with minimal (to no) material removal in the horizontal direction (here, x-direction and/or y-direction). In some embodiments, first spacer etch is a dry etch process, such as an RIE process, that uses a fluorine-containing etch gas to remove the material of spacer layer **240** (e.g., silicon nitride) at a higher rate than the materials of gate stacks **230A**, **230B** (e.g., oxide hard mask or polysilicon), isolation feature **225** (e.g., silicon oxide), and/or fins **222A**, **222B** (e.g., silicon) (i.e., the etchant has a high etch selectivity with respect to silicon nitride). In some embodiments, first spacer etch is configured to generate a fluorine-containing plasma from the fluorine-containing gas, such that first spacer etch removes portions of spacer layer **240** using plasma-excited fluorine-containing species. The fluorine-containing etch gas includes fluorine (F.sub.2), fluoromethane (e.g., CH.sub.3F), difluoromethane (e.g., CH.sub.2F.sub.2), trifluoromethane (e.g., CHF.sub.3), tetrafluoromethane (e.g., CF.sub.4), hexafluoroethane (e.g.,

C.sub.2F.sub.6), sulfur hexafluoride (e.g., SF.sub.6), nitrogen trifluoride (e.g., NF.sub.3), other fluorine-containing etchant, or combinations thereof. First spacer etch can, alternatively or additionally, use a hydrogen-containing etch gas (for example, H.sub.2 and/or CH.sub.4), a nitrogen-containing etch gas (for example, N.sub.2 and/or NH.sub.3), a chlorine-containing etch gas (for example, Cl.sub.2, CHCl.sub.3, CCl.sub.4, and/or BCl.sub.3), an oxygen-containing etch gas (for example, O.sub.2), a bromine-containing etch gas (for example, HBr and/or CHBr.sub.3), an iodine-containing etch gas, other suitable etch gas, or combinations thereof. First spacer etch may be configured to generate a plasma from any of the etch gases disclosed herein, such that first spacer etch uses plasma-excited species for etching. In some embodiments, a carrier gas is used to deliver the fluorine-containing etch gas and/or other etch gas. The carrier gas may be an inert gas, such as an argon-containing gas, a helium-containing gas, a xenon-containing gas, other suitable inert gas, or combinations thereof. In some embodiments, first spacer etch is a wet, anisotropic etch process.

[0030] Various etch parameters of first spacer etch can be tuned to achieve selective and anisotropic etching of spacer layer **240**, such as etch gas composition, carrier gas composition, etch gas flow rate, carrier gas flow rate, etch time, etch pressure, etch temperature, source power, radio frequency (RF) and/or direct current (DC) bias voltage, RF and/or DC bias power, other suitable etch parameters, or combinations thereof. In some embodiments, first spacer etch utilizes an etch gas that includes CF.sub.4 to selectively etch spacer layer **240**, where a flow rate of CF.sub.4, an etch time, an etch temperature, an etch pressure, and/or an RF power may be tuned to achieve desired etch selectivity. In some embodiments, a flow rate of CF.sub.4 is about 200 sccm to about 300 sccm. In some embodiments, first spacer etch utilizes an etch gas that includes CF.sub.4 and HBr, where a ratio of CF.sub.4 to HBr, an etch time, an etch temperature, an etch pressure, and/or an RF power may be tuned to achieve desired etch selectivity. In some embodiments, a ratio of a flow rate of CF.sub.4 to a flow rate of HBr is about 1:2 to about 2:1. In some embodiments, a flow rate of CF.sub.4 is about 200 sccm to about 300 sccm. In some embodiments, a flow rate of HBr is about 200 sccm to about 300 sccm. In some embodiments, a duration of first spacer etch is about 5 seconds to about 50 seconds. In some embodiments, an RF power used for generating a fluorine-containing plasma for first spacer etch is about 400 W to about 500 W. In some embodiments, a pressure maintained in a process chamber during first spacer etch is about 10 mTorr to about 50 mTorr. In some embodiments, a temperature maintained in the process chamber during first spacer etch is about 20° C. to about 70° C. In some embodiments, the various etch parameters of first spacer etch are tuned to achieve desired etching of fins **222A**, **222B**. In some embodiments, the various etch parameters of first spacer etch are tuned to achieve a desired thickness and/or a desired profile for gate spacers **242** and/or fin spacers **244**.

[0031] Turning to FIG. **3D** and FIG. **4D**, second spacer etch is performed to remove any remaining material of spacer layer **240** (e.g., residue) along horizontal surfaces of FinFET device **200**, such as top surfaces of fins **222A**, **222B**, top surfaces of isolation feature **225**, and top surfaces of gate stacks **230A**, **230B**. For example, second spacer etch selectively removes any remaining material of spacer layer **240** with respect to gate stacks **230A**, **230B**, isolation feature **225**, and/or fins **222A**, **222B**. In other words, second spacer etch substantially removes spacer layer **240** but does not remove, or does not substantially remove, gate stacks **230A**, **230B**, isolation feature **225**, and/or fins **222A**, **222B**. Second spacer etch may also remove portions of fins **222A**, **222B** in S/D regions to begin and/or continue formation of source/drain recesses **250**. For example, source/drain recesses **250** have depth d2 after second spacer etch, where depth d2 is greater than depth d1. Second spacer etch may also be used to adjust thickness t1 and/or a profile of gate spacers **242** and/or a thickness and/or a profile of fin spacers **244**.

[0032] Second spacer etch is an anisotropic etch process. For example, second spacer etch has a vertical etch rate that is greater than a horizontal etch rate (in some embodiments, the horizontal etch rate equals zero). Second spacer etch thus removes material in substantially the vertical

direction (here, z-direction) with minimal (to no) material removal in the horizontal direction (here, x-direction and/or y-direction). In some embodiments, an etch selectivity of silicon nitride (i.e., spacer layer **240**) to silicon (i.e., fins **222A**, **222B**) of second spacer etch is greater than an etch selectivity of silicon nitride to silicon of first spacer etch. In such embodiments, fins **222A**, **222B** may be etched more by first spacer etch than second spacer etch, such that a thickness (or depth) of fins **222A**, **222B** removed by first spacer etch is more than a thickness (or depth) of fins **222A**, **222B** removed by second spacer etch. In some embodiments, second spacer etch is a dry etch process, such as an RIE process, that uses a fluorine-and-hydrogen containing etch gas to remove the material of spacer layer **240** (e.g., silicon nitride) at a higher rate than the materials of gate stacks **230A**, **230B** (e.g., oxide hard mask or polysilicon), isolation feature **225** (e.g., silicon oxide), and/or fins **222A**, **222B** (e.g., silicon) (i.e., the etchant has a high etch selectivity with respect to silicon nitride). In some embodiments, second spacer etch is configured to generate a fluorine-and-hydrogen containing plasma from the fluorine-and-hydrogen containing gas, such that second spacer etch removes portions of spacer layer **240** using plasma-excited fluorine-and-hydrogen containing species. The fluorine-and-hydrogen containing etch gas includes fluoromethane (e.g., CH_3F), difluoromethane (e.g., CH_2F_2), trifluoromethane (e.g., CHF_3), other fluorine-and-hydrogen containing etchant, or combinations thereof. Second spacer etch can, alternatively or additionally, use a hydrogen-containing etch gas (for example, H_2 and/or CH_4), a nitrogen-containing etch gas (for example, N_2 and/or NH_3), a chlorine-containing etch gas (for example, Cl_2 , CHCl_3 , CCl_4 , and/or BCl_3), an oxygen-containing etch gas (for example, O_2), a bromine-containing etch gas (for example, HBr and/or CH_3Br), an iodine-containing etch gas, a fluorine-containing etch gas (for example, F_2 , CF_4 , C_2F_6 , SF_6 , and/or NF_3), other suitable etch gas, or combinations thereof. Second spacer etch may be configured to generate a plasma from any of the etch gases disclosed herein, such that second spacer etch uses plasma-excited species for etching. In some embodiments, a carrier gas is used to deliver the fluorine-and-hydrogen containing etch gas and/or other etch gas. The carrier gas may be an inert gas, such as an argon-containing gas, a helium-containing gas, a xenon-containing gas, other suitable inert gas, or combinations thereof. In some embodiments, second spacer etch is a wet, anisotropic etch process.

[0033] Various etch parameters of second spacer etch can be tuned to achieve selective and anisotropic etching of spacer layer **240**, such as etch gas composition, carrier gas composition, etch gas flow rate, carrier gas flow rate, etch time, etch pressure, etch temperature, source power, RF and/or DC bias voltage, RF and/or DC bias power, other suitable etch parameters, or combinations thereof. In some embodiments, second spacer etch utilizes an etch gas that includes CH_3F and O_2 , where a ratio of CH_3F to O_2 , an etch time, an etch temperature, an etch pressure, and/or an RF power may be tuned to achieve desired etch selectively. In some embodiments, a ratio of a flow rate of CH_3F to a flow rate of O_2 is about 1:1 to about 4:1. In some embodiments, a flow rate of CH_3F is about 100 sccm to about 200 sccm. In some embodiments, a flow rate of O_2 is about 50 sccm to about 150 sccm. In some embodiments, a duration of second spacer etch is about 5 seconds to about 50 seconds. In some embodiments, an RF power used for generating a hydrogen-and-fluorine containing plasma for second spacer etch is about 400 W to about 500 W. In some embodiments, a pressure maintained in a process chamber during second spacer etch is about 10 mTorr to about 50 mTorr. In some embodiments, a temperature maintained in the process chamber during second spacer etch is about 20° C. to about 70° C. In some embodiments, the various etch parameters of second spacer etch are tuned to achieve desired etching of fins **222A**, **222B** in S/D regions **250**. In some embodiments, the various etch parameters of first spacer etch are tuned to achieve a desired thickness and/or a desired profile for gate spacers **242** and/or fin spacers **244**.

[0034] Processing then proceeds, as depicted in FIGS. 3E-3G and FIGS. 4E-4G, with performing a three-step source/drain etch process to form source/drain recesses **250** in S/D regions of fins **222A**,

222B. Source/drain recesses **250** extend in fins **222A**, **222B** to a target depth TD after the three-step source/drain etch process. In some embodiments, target depth TD is about 80% to about 120% of fin height FH (e.g., about 120% of fin height FH). In some embodiments, target depth TD is about 30 nm to about 80 nm. When fabricating gate spacers **242** begins formation of source/drain recesses **250**, such as in the depicted embodiment, three-step source/drain etch process completes formation of source/drain recesses **250**. In the depicted embodiment, three-step source/drain etch process includes a first anisotropic source/drain etch (FIG. 3E and FIG. 4E), an isotropic source/drain etch (FIG. 3F and FIG. 4F), and a second anisotropic source/drain etch (FIG. 3G and FIG. 4G). Three-step source/drain etch process provides precise control of a location of tips of epitaxial source/drain features (in particular, a depth of the tips in fins **222A**, **222B** and a proximity (distance) of the tips to the channel regions of fins **222A**, **222B**) and a distance between a bottom portion of the epitaxial source/drain features and the channel region and/or the gate stack. For example, first anisotropic source/drain etch and isotropic source/drain etch are tuned to achieve a location of tips of epitaxial source/drain features before extending source/drain recesses to target depth TD. In some embodiments, the location of tips of epitaxial source/drain features is defined before the source/drain recesses reach a depth that is greater than about 80% of fin height FH (i.e., a depth of source/drain recesses after first anisotropic source/drain etch and isotropic source/drain etch is less than or equal to about 80% of fin height FH). Extending the source/drain recesses to target depth after defining the location of tips of epitaxial source/drain features pulls back a proximity of a bottom portion of the epitaxial source/drain features from gate footing GF, which reduces parasitic capacitance between the epitaxial source/drain features and the gate stack.

[0035] Turning to FIG. 3E and FIG. 4E, first anisotropic source/drain etch extends source/drain recesses **250** to a depth d3 in S/D regions of fins **222A**, **222B**. Depth d3 is measured between respective top surfaces of fins **222A**, **222B** and respective portions of fins **222A**, **222B** defining bottommost surfaces of source/drain recesses **250** after first anisotropic source/drain etch. Depth d3 is less than target depth TD and is selected based on a desired tip depth of source/drain recesses **250** (i.e., a depth of a tip along fin height FH). In some embodiments, depth d3 is about 10% to about 80% of fin height FH depending on desired tip depth. In some embodiments, after the three-step source/drain recess etch, a location of tip depth along fin height FH is about 40% to about 60% of depth d3 (i.e., $0.4d3 \leq \text{tip depth} \leq 0.6d3$). In such embodiments, where desired tip depth is about 20% to about 30% of fin height FH, first anisotropic source/drain etch is performed until depth d3 is about 50% of fin height FH. In some embodiments, depth d3 is selected so the tip depth is less than about 50% of fin height FH. For example, depth d3 is selected to achieve a tip depth of about 10% to about 30% of fin height FH. In such embodiments, depth d3 can be about 25% to about 50% of fin height FH. In some embodiments, depth d3 is about 10 nm to about 60 nm. Where first spacer etch and second spacer etch begin formation of source/drain recesses **250**, such as in the depicted embodiment, depth d3 is greater than depth d1, depth d2, and a sum of depth d1 and depth d2.

[0036] First anisotropic source/drain etch is controlled to produce source/drain recesses **250** having U-shaped profiles in the Y-Z plane. For example, after first anisotropic source/drain etch, source/drain recesses **250** are defined by a sidewall segment A, a sidewall segment B, and a bottom segment C that extends between sidewall segment A and sidewall segment B. Sidewall segment A is substantially linear, sidewall segment B is substantially linear, and bottom segment C is curvilinear. In the depicted embodiment, sidewall segment A and sidewall segment B extend substantially parallel to one another along the z-direction. Source/drain recesses **250** have a width w1. In some embodiments, width w1 is about 10 nm to about 35 nm. In furtherance of the depicted embodiment, source/drain recesses **250** extend under gate spacers **242**, resulting in sidewall segment A being offset a distance along the y-direction from an outer sidewall OS of a respective gate spacer **242** along gate stack **230A** and sidewall segment B being offset a distance along the y-direction from an outer sidewall OS of a respective gate spacer **242** along gate stack **230B**. In such

embodiments, width w_1 is greater than a spacing S between outer sidewalls OSs of gate spacers **242**. In some embodiments, source/drain recesses **250** do not extend under gate spacers **242**. For example, sidewall segment A may be substantially aligned along the z-direction with outer sidewall OS of respective gate spacer **242** along gate stack **230A** and sidewall segment B may be substantially aligned along the z-direction with outer sidewall OS of respective gate spacer **242** along gate stack **230B**. In such embodiments, width w_1 is about equal to spacing S between outer sidewalls OSs of gate spacers **242**. The present disclosure also contemplates embodiments where sidewall segment A and sidewall segment B are substantially linear having slight slope in the Y-Z plane, such as sidewall segment A having a slight negative slope and sidewall segment B having a slight positive slope.

[0037] First anisotropic source/drain etch selectively removes fins **222A**, **222B** with respect to gate spacers **242**, fin spacers **244**, gate stacks **230A**, **230B**, and/or isolation feature **225**. In other words, first anisotropic source/drain etch substantially removes fins **222A**, **222B** but does not remove, or does not substantially remove, gate spacers **242**, fin spacers **244**, gate stacks **230A**, **230B**, and/or isolation feature **225**. First anisotropic source/drain etch has a vertical etch rate that is greater than a horizontal etch rate (in some embodiments, the horizontal etch rate is zero). First anisotropic source/drain etch thus removes portions of fins **222A**, **222B** in substantially the vertical direction (here, z-direction) with minimal (to no) removal of material in the horizontal direction (here, x-direction and/or y-direction). For example, first anisotropic source/drain etch has a ratio of a vertical etch rate (VER) to a horizontal etch rate (HER) that is greater than about 20 (e.g., $20 \leq \text{VER}/\text{HER} \leq 80$). In some embodiments, first anisotropic source/drain etch is controlled to exhibit a desired degree of anisotropy (A), which is defined as one minus a ratio of HER to VER (i.e., $A = 1 - \text{HER}/\text{VER}$). In some embodiments, a degree of anisotropy is about 0.95 to about 1. In some embodiments, vertical etching refers to etching downward in a direction perpendicular to top surfaces of fins **222A**, **222B**, and horizontal etching refers to etching in directions parallel to top surfaces of fins **222A**, **222B**. In some embodiments, first anisotropic source/drain etch is a dry etch process, such as an RIE process, that uses a bromine-containing etch gas to remove the material of fins **222A**, **222B** (e.g., silicon) at a higher rate than the materials of gate spacers **242** (e.g., silicon nitride), fin spacers **244** (e.g., silicon nitride), gate stacks **230A**, **230B** (e.g., oxide hard mask or polysilicon), and/or isolation feature **225** (e.g., silicon oxide) (i.e., the etchant has a high etch selectivity with respect to silicon). The bromine-containing etch gas includes bromine (Br), hydrogen bromide (HBr), bromomethane (CH_3Br), dibromomethane (CH_2Br_2), bromoform (CHBr_3), other bromine-containing etchant, or combinations thereof. In some embodiments, first anisotropic source/drain etch is configured to generate a bromine-containing plasma from the bromine-containing etch gas, such that first anisotropic source/drain etch removes portions of fins **222A**, **222B** in S/D regions using plasma-excited bromine-containing species. First anisotropic source/drain etch can, alternatively or additionally, use a hydrogen-containing etch gas (for example, H_2 and/or CH_4), a nitrogen-containing etch gas (for example, N_2 and/or NH_3), a chlorine-containing etch gas (for example, Cl_2 , CHCl_3 , CCl_4 , and/or BCl_3), an oxygen-containing etch gas (for example, O_2), a fluorine-containing etch gas (for example, F_2 , CHF_3 , CH_2F_2 , CHF_3 , CF_4 , C_2F_6), an iodine-containing etch gas, other suitable etch gas, or combinations thereof. In some embodiments, a carrier gas is used to deliver the etch gas. The carrier gas may be an inert gas, such as an argon-containing gas, a helium-containing gas, a xenon-containing gas, other suitable inert gas, or combinations thereof. In some embodiments, first anisotropic source/drain etch is a wet, anisotropic etch process.

[0038] Various etch parameters of first anisotropic source/drain etch are tuned to control selective and anisotropic etching of fins **222A**, **222B** to produce desired depth d_3 and U-shaped profiles of source/drain recesses **250**, such as etch gas composition, carrier gas composition, etch gas flow rate, carrier gas flow rate, etch time, etch pressure, etch temperature, source power, RF and/or DC

bias voltage, RF and/or DC bias power, other suitable etch parameters, or combinations thereof. In some embodiments, first anisotropic source/drain etch utilizes an etch gas that includes HBr, where a flow rate of HBr, an etch time, an etch temperature, an etch pressure, and/or an RF power may be tuned to achieve desired etch selectivity and/or desired source/drain recess profile. In such embodiments, a flow rate of HBr is about 100 sccm to about 200 sccm. In some embodiments, a duration of first anisotropic source/drain etch is about 30 seconds to about 80 seconds. In some embodiments, RF power used for generating a bromine-containing plasma is about 300 W to about 400 W. In some embodiments, a pressure maintained in a process chamber during first anisotropic source/drain etch is about 5 mTorr to about 30 mTorr. In some embodiments, a temperature maintained in the process chamber during first anisotropic source/drain etch is about 20° C. to about 70° C. In some embodiments, first anisotropic source/drain etch utilizes an etch gas that includes Cl.sub.2, where a flow rate of Cl.sub.2, an etch time, an etch temperature, an etch pressure, and/or an RF power may be tuned to achieve desired etch selectivity and/or desired source/drain recess profile.

[0039] Turning to FIG. 3F and FIG. 4F, isotropic source/drain etch is performed to widen source/drain recesses 250 and control a tip proximity of tips source/drain recesses 250. Isotropic source/drain etch is controlled to produce source/drain recesses 250 having bowl-shaped profiles in the Y-Z plane. For example, after isotropic source/drain etch, source/drain recesses 250 are defined by a sidewall segment A', a sidewall segment B', and a bottom segment C' that extends between sidewall segment A' and sidewall segment B'. Sidewall segment A' is curvilinear, sidewall segment B' is curvilinear, and bottom segment C' is curvilinear. Sidewall segment A' extends away from a top surface of fins 222A, 222B towards gate stack 230A to a source/drain tip v1, where sidewall segment A' meets bottom segment C'. Sidewall segment B' extends away from a top surface of fins 222A, 222B towards gate stack 230B to a source/drain tip v2, where sidewall segment B' meets bottom segment C'. Such configuration results in source/drain recesses 250 having a wide central portion disposed between narrower end portions, where the narrower end portions have widths that decreasingly taper along the z-direction away from the wide central portion. Source/drain recesses 250 thus have a width w2 that varies along fin height FH, where width w2 equals a maximum width of source/drain recesses 250 between source/drain tip v1 and source/drain tip v2. Source/drain tips v1, v2 have an initial tip depth dti, where tip depth generally refers to a vertical distance between a gate stack (and/or top surface of a fin) and a portion of a source/drain feature that is closest to the gate stack (i.e., a tip). In some embodiments, initial tip depth dti is about 10% to about 40% of fin height FH. Source/drain tips v1, v2 also have an initial tip proximity pi, where tip proximity generally refers to a lateral distance between a sidewall of a gate stack and a portion of a source/drain feature that is closest to the sidewall of the gate stack (i.e., a distance between source/drain tips v1, v2 and a respective channel region). In some embodiments, initial tip proximity pi is about 60% to about 90% of thickness t1 of gate spacers 242 (e.g., $0.6t_1 \leq p \leq 0.9t_1$). In some embodiments, initial tip proximity pi is about 5 nm to about 8 nm. Isotropic source/drain etch may also deepen source/drain recesses 250, for example, extending source/drain recesses 250 to a depth d4 that is greater than depth d3 and less than target depth TD. Depth d4 is measured between respective top surfaces of fins 222A, 222B and respective portions of fins 222A, 222B defining bottommost surfaces of source/drain recesses 250 after isotropic source/drain etch. In some embodiments, depth d4 is about 10% to about 80% of fin height FH, such as about 50% of fin height FH. In some embodiments, depth d4 is about 10 nm to about 70 nm.

[0040] Isotropic source/drain etch selectively removes fins 222A, 222B with respect to gate spacers 242, fin spacers 244, gate stacks 230A, 230B, and/or isolation feature 225. In other words, isotropic source/drain etch substantially removes fins 222A, 222B but does not remove, or does not substantially remove, gate spacers 242, fin spacers 244, gate stacks 230A, 230B, and/or isolation feature 225. In contrast to first anisotropic source/drain etch, isotropic source/drain etch etches in multiple directions (in some embodiments, the vertical etch rate is the same as the horizontal etch

rate). Isotropic source/drain etch thus removes portions of fins **222A**, **222B** in all directions (here, z-direction, y-direction, and x-direction). For example, isotropic source/drain etch has a ratio of VER to HER that is about 1. In some embodiments, isotropic source/drain etch has a ratio of VER to HER that is about 1 to about 5 (e.g., $1 \leq \text{VER/HER} \leq 5$). In some embodiments, isotropic source/drain etch is controlled to exhibit a degree of anisotropy that is less than the degree of anisotropy of first anisotropic source/drain etch. For example, a degree of anisotropy of isotropic source/drain etch is less than about 0.8 (in some embodiments, 0). In some embodiments, isotropic source/drain etch is a dry etch process, such as an RIE process, that uses a fluorine-and-chlorine containing etch gas to remove the material of fins **222A**, **222B** (e.g., silicon) at a higher rate than the materials of gate spacers **242** (e.g., silicon nitride), fin spacers **244** (e.g., silicon nitride), gate stacks **230A**, **230B** (e.g., oxide hard mask or polysilicon), and/or isolation feature **225** (e.g., silicon oxide) (i.e., the etchant has a high etch selectivity with respect to silicon). The fluorine-and-chlorine containing etch gas includes a fluorine-containing etchant (e.g., $\text{F}_{\text{sub.2}}$, $\text{CH}_{\text{sub.3}}\text{F}$, $\text{CH}_{\text{sub.2}}\text{F}_{\text{sub.2}}$, $\text{CHF}_{\text{sub.3}}$, $\text{CF}_{\text{sub.4}}$, $\text{C}_{\text{sub.2}}\text{F}_{\text{sub.6}}$, $\text{SF}_{\text{sub.6}}$, $\text{NF}_{\text{sub.3}}$, other fluorine-containing etchant, or combinations thereof) and a chlorine-containing etchant (e.g., for example, $\text{Cl}_{\text{sub.2}}$, $\text{CHCl}_{\text{sub.3}}$, $\text{CCl}_{\text{sub.4}}$, $\text{BCl}_{\text{sub.3}}$, other chlorine-containing etchant, or combinations thereof). In some embodiments, isotropic source/drain etch is configured to generate a fluorine-and-chlorine containing plasma from the fluorine-and-chlorine containing etch gas, such that isotropic source/drain etch removes portions of fins **222A**, **222B** using plasma-excited fluorine-containing species and/or plasma-excited chlorine-containing species. Isotropic source/drain etch can, alternatively or additionally, use a hydrogen-containing etch gas (for example, $\text{H}_{\text{sub.2}}$ and/or $\text{CH}_{\text{sub.4}}$), a nitrogen-containing etch gas (for example, $\text{N}_{\text{sub.2}}$ and/or $\text{NH}_{\text{sub.3}}$), an oxygen-containing etch gas (for example, $\text{O}_{\text{sub.2}}$), a bromine-containing etch gas (for example, HBr and/or $\text{CHBr}_{\text{sub.3}}$), an iodine-containing etch gas, other suitable etch gas, or combinations thereof. In some embodiments, a carrier gas is used to deliver the etch gas. The carrier gas may be an inert gas, such as an argon-containing gas, a helium-containing gas, a xenon-containing gas, other suitable inert gas, or combinations thereof. In some embodiments, isotropic source/drain etch is a wet etch process that uses an etch solution that includes nitric acid ($\text{HNO}_{\text{sub.3}}$) and hydrofluoric acid (HF). In some embodiments, the etch solution further includes acetic acid ($\text{CH}_{\text{sub.3}}\text{COOH}$). The present disclosure contemplates other etch solutions for isotopically and selectively wet etching.

[0041] Various etch parameters of isotropic source/drain etch can be tuned to control selective and isotropic etching of fins **222A**, **222B** to produce desired tip proximity pi , desired tip depth di , and desired bowl-shaped profiles of source/drain recesses **250**, such as etch gas composition, carrier gas composition, etch gas flow rate, carrier gas flow rate, etch time, etch pressure, etch temperature, source power, RF and/or DC bias voltage, RF and/or DC bias power, other suitable etch parameters, or combinations thereof. In some embodiments, isotropic source/drain etch utilizes an etch gas that includes $\text{NF}_{\text{sub.3}}$ and $\text{Cl}_{\text{sub.2}}$ to selectively etch fins **222A**, **222B**, where a ratio of a flow rate of $\text{NF}_{\text{sub.3}}$ to a flow rate of $\text{Cl}_{\text{sub.2}}$, a flow rate of $\text{NF}_{\text{sub.3}}$, a flow rate of $\text{Cl}_{\text{sub.2}}$, an etch time, an etch temperature, an etch pressure, and/or an RF power may be tuned to achieve desired etch selectivity and/or desired source/drain recess profile. In some embodiments, a ratio of a flow rate of $\text{NF}_{\text{sub.3}}$ to a flow rate of $\text{Cl}_{\text{sub.2}}$ is about 1:7 to about 1:1. In some embodiments, a flow rate of $\text{NF}_{\text{sub.3}}$ is about 100 sccm to about 300 sccm. In some embodiments, a flow rate of $\text{Cl}_{\text{sub.2}}$ is about 100 sccm to about 200 sccm. In some embodiments, a duration of isotropic source/drain etch is about 5 seconds to about 50 seconds. In some embodiments, RF power used for generating a fluorine-and-oxygen containing plasma for isotropic source/drain etch is about 300 W to about 400 W. In some embodiments, a pressure maintained in a process chamber during isotropic source/drain etch is about 5 mTorr to about 50 mTorr. In some embodiments, a temperature maintained in the process chamber during isotropic source/drain etch is about 20° C. to about 70° C.

[0042] Turning to FIG. 3G and FIG. 4G, second anisotropic source/drain etch extends source/drain recesses **250** to a depth d5, which is substantially equal to target depth TD. Depth d5 is measured between respective top surfaces of fins **222A**, **222B** and respective portions of fins **222A**, **222B** defining bottommost surfaces of source/drain recesses **250** after second anisotropic source/drain etch. In some embodiments, depth d5 is about 80% to about 120% of fin height FH (for example, about 120% of fin height FH). In some embodiments, depth d5 is about 30 nm to about 80 nm. Second anisotropic source/drain etch is controlled to produce source/drain recesses **250** having a profile in the Y-Z plane that resembles a light portion of a candle-shaped light bulb. For example, after second anisotropic source/drain etch, source/drain recesses **250** are defined by a sidewall segment A'', a sidewall segment B'', and a bottom segment C'' that extends between sidewall segment A'' and sidewall segment B''. Sidewall segment A'' is substantially linear having positive slope in the Y-Z plane, sidewall segment B'' is substantially linear having negative slope in the Y-Z plane, and bottom segment C'' is curvilinear. Accordingly, sidewall segment A'' extends away from a top surface of fins **222A**, **222B** towards gate stack **230A** to source/drain tip v1, where sidewall segment A'' meets bottom segment C'', and sidewall segment B'' extends away from a top surface of fins **222A**, **222B** towards gate stack **230B** to source/drain tip v2, where sidewall segment B'' meets bottom segment C''. Such configuration results in source/drain recesses **250** maintaining a maximum width between source/drain tip v1 and source/drain tip v2. A first portion of source/drain recesses **250**, defined between sidewall segment A'' and sidewall segment B'', has a width w3 that increases from top surfaces of fins **222A**, **222B** to the maximum width of source/drain recesses **250**. A second portion of source/drain recesses **250**, defined between sidewall portions of bottom segment C'', has a width w4 that decreases from the maximum width of source/drain recesses **250** to a width less than the maximum width (in some embodiments, 0). In the depicted embodiment, bottom segment C'' has a substantially V-shaped blunt-tip profile. For example, bottom segment C'' have substantially linear sidewalls, where the substantially linear sidewalls have opposite slope and taper into a curved bottom instead of a point. The present disclosure also contemplates embodiments where sidewall segment A'' and/or sidewall segment B'' are curvilinear. For example, sidewall segment A'' and/or sidewall segment B'' may initially curve towards one another from top surfaces of fins **222A**, fins **222B** and then curve away from one another towards source/drain tips v1, v2, respectively, thereby forming convex surfaces that define the first portion of source/drain recesses **250** therebetween.

[0043] Source/drain recesses **250** have profiles that improve performance of FinFET device **200** compared to profiles of conventional source/drain recesses **250'**, which result from source/drain fabrication techniques that customize a location of source/drain tips after reaching target depth TD. For example, a location of source/drain tips v1, v2 is closer to gate stacks **230A**, **230B** (and thus closer to channel regions of FinFET device **200**) compared to a location of source/drain tips z1, z2 of conventional source/drain recesses **250'**. Source/drain tips v1, v2 have a tip depth dt and a tip proximity pt that optimize performance of FinFET device **200**. In some embodiments, tip depth dt is about 10% to about 40% of fin height FH, and tip proximity pt is about 50% to about 90% of thickness t1 of gate spacers **242**, which has been observed to reduce parasitic resistance between subsequently formed source/drain features and subsequently formed metal gates. In some embodiments, tip depth dt is about 10 nm to about 30 nm (for example, about 20 nm), and tip proximity pt is about 5 nm to about 8 nm (for example, about 6 nm). Tip depth dt greater than about 40% of fin height FH (in some embodiments, greater than about 30 nm) has been observed to pull a bottom of source/drain recesses **250** towards gate stacks **230A**, **230B** during processing, undesirably decreasing proximity footing pf and increasing parasitic capacitance between subsequently formed source/drain features and subsequently formed metal gates, as described herein. Tip depth dt less than about 10% of fin height FH (in some embodiments, less than about 10 nm) may increase parasitic resistance. Tip proximity pt less than 50% of thickness t1 of gate spacers **242** (in some embodiments, about 8 nm) has been observed to increase resistance between

subsequently formed source/drain features and the channel region, reducing carrier mobility, while tip proximity pt greater than 90% of thickness t_1 of gate spacers **242** (in some embodiments, about 5 nm) has been observed to push subsequently formed source/drain features too close to the channel region, which can lead to undesired short channel effects and/or off-state current leakage. Tip depth dt may be greater than, less than, or the same as initial tip depth dt_i . In some embodiments, second anisotropic source/drain etch may increase, decrease, and/or have no effect on tip depth depending on the various etch parameters of second anisotropic source/drain etch. In the depicted embodiment, second anisotropic source/drain etch results in tip depth dt being greater than initial tip depth dt_i . In some embodiments, second anisotropic source/drain etch may increase and/or have no effect on tip proximity depending on the various etch parameters of second anisotropic source/drain etch. In the depicted embodiment, second anisotropic source/drain etch is tuned to ensure tip proximity pt is substantially the same as initial tip proximity pt_i . After second source/drain anisotropic etch, a proximity of a bottom portion of source/drain recesses **250** (in particular, bottom segment C'') to gate footing GF (referred to as proximity footing pf) also improves performance of FinFET device **200**. For example, because a location of source/drain tips v_1, v_2 is substantially defined by first source/drain anisotropic etch and isotropic source/drain etch before source/drain recesses **250** reach target depth TD, proximity footing pf of source/drain recesses **250** is greater than a proximity footing pf of conventional source/drain recesses **250'**, which results from conventional source/drain fabrication techniques. In some embodiments, proximity footing pf is greater than or equal to about 8 nm, in particular, at depths greater than about 80% of fin height FH. Increasing proximity footing (for example, greater than or equal to about 8 nm) at such depths (for example, greater than about 80% of fin height FH) reduces parasitic capacitance between subsequently formed source/drain features and gate footing GF, which reduces DIBL and improves short channel control of FinFET device **200** compared to FinFET devices having source/drain recesses with profiles similar to conventional source/drain recesses **250'**, which have been observed to have proximity footing that is less than 8 nm at depths greater than about 80% of fin height FH. A tapering of bottom segment C'' from source/drain tips v_1, v_2 to target depth TD is also greater than a tapering of a bottom segment from source/drain tips z_1, z_2 to target depth TD, such that a distance between bottom segment C'' of source/drain recesses **250** and gate stacks **230A, 230B** is greater than a distance between the bottom segment of conventional source/drain recesses **250'** and gate stacks **230A, 230B**. This increase in distance compared to conventional source/drain recesses **250'** can further reduce parasitic capacitance. In some embodiments, the distance between bottom segment C'' of source/drain recesses **250** and gate stacks **230A, 230B** is greater than or equal to about 6 nm at depths greater than about 50% of fin height FH. When a distance between a bottom segment of source/drain recesses and gate stacks **230A, 230B** is less than about 6 nm, such as the distances observed between bottom segment C' of conventional source/drain recesses **250'** and gate stacks **230A, 230B**, parasitic capacitance between lower portions of subsequently formed source/drain features and subsequently formed metal gates increases, thereby increasing DIBL and degrading short channel control.

[0044] Second anisotropic source/drain etch selectively removes fins **222A, 222B** with respect to gate spacers **242**, fin spacers **244**, gate stacks **230A, 230B**, and/or isolation feature **225**. In other words, second anisotropic source/drain etch substantially removes fins **222A, 222B** but does not remove, or does not substantially remove, gate spacers **242**, fin spacers **244**, gate stacks **230A, 230B**, and/or isolation feature **225**. Second anisotropic source/drain etch has a vertical etch rate that is greater than a horizontal etch rate (in some embodiments, the horizontal etch rate is zero), such that second anisotropic source/drain etch removes portions of fins **222A, 222B** in substantially the vertical direction (here, z -direction) with minimal (to no) removal of material in the horizontal direction (here, x -direction and/or y -direction). For example, second anisotropic source/drain etch has a ratio of VER to HER that is greater than about 20 (e.g., $20 \leq \text{VER/HER} \leq 80$). In some embodiments, second anisotropic source/drain etch is controlled to exhibit a degree of anisotropy

that is substantially the same as the degree of anisotropy of first anisotropic source/drain etch. For example, a degree of anisotropy of second anisotropic source/drain etch is about 0.95 to about 1. In some embodiments, second anisotropic source/drain etch is a dry etch process, such as an RIE process, that uses a bromine-containing etch gas to remove the material of fins **222A**, **222B** (e.g., silicon) at a higher rate than the materials of gate spacers **242** (e.g., silicon nitride), fin spacers **244** (e.g., silicon nitride), gate stacks **230A**, **230B** (e.g., oxide hard mask or polysilicon), and/or isolation feature **225** (e.g., silicon oxide) (i.e., the etchant has a high etch selectivity with respect to silicon). The bromine-containing etch gas includes Br, HBr, CH₃Br, CH₂Br₂, CHBr₃, other bromine-containing etchant, or combinations thereof. In some embodiments, second anisotropic source/drain etch is configured to generate a bromine-containing plasma from the bromine-containing etch gas, such that second anisotropic source/drain etch removes portions of fins **222A**, **222B** in S/D regions using plasma-excited bromine-containing species. Second anisotropic source/drain etch can, alternatively or additionally, use a hydrogen-containing etch gas (for example, H₂ and/or CH₄), a nitrogen-containing etch gas (for example, N₂ and/or NH₃), a chlorine-containing etch gas (for example, Cl₂, CHCl₃, CCl₄, and/or BCl₃), an oxygen-containing etch gas (for example, O₂), a fluorine-containing etch gas (for example, F₂, CH₃F, CH₂F₂, CHF₃, CF₄, C₂F₆), an iodine-containing etch gas, other suitable etch gas, or combinations thereof. In some embodiments, a carrier gas is used to deliver the etch gas. The carrier gas may be an inert gas, such as an argon-containing gas, a helium-containing gas, a xenon-containing gas, other suitable inert gas, or combinations thereof.

[0045] Various etch parameters of second anisotropic source/drain etch can be tuned to control selective and anisotropic etching of fins **222A**, **222B** to ensure source/drain recesses **250** reach target depth TD and a profile in the Y-Z plane that resembles a light portion of a candle-shaped light bulb, such as etch gas composition, carrier gas composition, etch gas flow rate, carrier gas flow rate, etch time, etch pressure, etch temperature, source power, RF and/or DC bias voltage, RF and/or DC bias power, other suitable etch parameters, or combinations thereof. In some embodiments, second anisotropic source/drain etch utilizes an etch gas that includes HBr. In such embodiments, a flow rate of HBr may be about 100 sccm to about 200 sccm. In some embodiments, a duration of second anisotropic source/drain etch is about 10 seconds to about 50 seconds. In some embodiments, RF power used for generating a bromine-containing plasma is about 300 W to about 400 W. In some embodiments, a pressure maintained in a process chamber during second anisotropic source/drain etch is about 5 mTorr to about 50 mTorr. In some embodiments, a temperature maintained in the process chamber during second anisotropic source/drain etch is about 20° C. to about 70° C. It is noted that second anisotropic source/drain etch is substantially the same as first anisotropic source/drain etch, except that a duration of second anisotropic source/drain etch is different (in some embodiments, greater) than first anisotropic source/drain etch to ensure source/drain recesses **250** extend to target depth TD after second anisotropic source/drain etch. In some embodiments, second anisotropic source/drain etch utilizes an etch gas that includes Cl₂, where a flow rate of Cl₂, an etch time, an etch temperature, an etch pressure, and/or an RF power may be tuned to achieve desired etch selectivity and/or desired source/drain recess profile.

[0046] In FIG. 3H and FIG. 4H, epitaxial source/drain features **260** are formed in source/drain recesses **250**. A first gate structure (including gate stack **230A** and gate spacers **242**) interposes respective epitaxial source/drain features **260**, such that a channel region is defined between the respective epitaxial source/drain features **260**. A second gate structure (including gate stack **230B** and gate spacers **242**) interposes respective epitaxial source/drain features **260**, such that a channel region is defined between the respective epitaxial source/drain features **260**. In some embodiments, the first gate structure and its respective epitaxial source/drain features **260** form a portion of a first FinFET, and the second gate structure and its respective epitaxial source/drain features **260** form a

portion of a second FinFET. In some embodiments, a deposition process is performed to fill source/drain recesses **250** with epitaxial semiconductor material, thereby forming epitaxial source/drain features **260**. For example, a semiconductor material is epitaxially grown from portions of fins **222A**, **222B** and/or substrate **210** defining source/drain recesses **250**. An epitaxy process can implement CVD deposition techniques (for example, vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), LPCVD, and/or PECVD), molecular beam epitaxy, other suitable SEG processes, or combinations thereof. The epitaxy process can use gaseous (for example, Si-containing gases, such as SiH₄, and/or Ge-containing gases, such as GeH₄) and/or liquid precursors, which interact with the composition of fins **222A**, **222B** and/or substrate **210**. Epitaxial source/drain features **260** are doped with n-type dopants and/or p-type dopants. In some embodiments, epitaxial source/drain features **260** are epitaxial layers including silicon and/or carbon, where the silicon-comprising epitaxial layers or the silicon-carbon-comprising epitaxial layers are doped with phosphorous, other n-type dopant, or combinations thereof. In some embodiments, epitaxial source/drain features **260** are epitaxial layers including silicon and germanium, where the silicon-and-germanium-comprising epitaxial layers are doped with boron, other p-type dopant, or combinations thereof. In some embodiments, epitaxial source/drain features **260** include materials and/or dopants that achieve desired tensile stress and/or compressive stress in the channel regions. In some embodiments, epitaxial source/drain features **260** are doped during deposition by adding impurities to a source material of the epitaxy process. In some embodiments, epitaxial source/drain features **260** are doped by an ion implantation process subsequent to a deposition process. In some embodiments, annealing processes are performed to activate dopants in epitaxial source/drain features **260** and/or other source/drain regions of FinFET device **200** (for example, HDD regions and/or LDD regions).

[0047] Epitaxial source/drain features **260** have candle light bulb-shaped profiles in the Y-Z plane. For example, epitaxial source/drain features **260** have a top portion **262A** (a portion of epitaxial source/drain features **260** disposed above top surfaces of fins **222A**, **222B** along the z-direction), a middle portion **262B** (a portion of epitaxial source/drain features **260** disposed between top surfaces of fins **222A**, **222B** and tips T), and a bottom portion **262C** (a portion of epitaxial source/drain features **260** disposed between tips T and target depth TD). Top portion **262A** resembles a base portion of a candle-shaped light bulb while middle portion **262B** and bottom portion **262C** combine to resemble a light portion of a candle-shaped light bulb (having, for example, a wide central portion disposed between two narrower end portions), which is configured to resemble a flame of a candle. In the depicted embodiment, top portion **262A** has a width w_t that is substantially the same along its thickness t_t , middle portion **262B** has a width w_m that increases along its thickness t_m from top surfaces of fins **222A**, **222B** to tips T (i.e., an outwardly tapering width along fin height FH towards target depth TD), and bottom portion **262C** has a width w_b that decreases along its thickness t_b from tips T to target depth TD (i.e., an inwardly tapering width along fin height FH toward target depth TD). A maximum width of epitaxial source/drain features **260** is defined between tips T. In some embodiments, maximum width is about 20 nm to about 35 nm. In some embodiments, as depicted, width w_t is substantially equal to spacing S between gate spacers **242**, such that top portion **262A** physically contacts outer sidewalls OSs of gate spacers **242**. In some embodiments, width w_t is less than spacing S, such that top portion **262A** does not physically contact or only partially physically contacts outer sidewalls OSs of gate spacers **242**. In furtherance of the depicted embodiment, bottom portion **262C** is defined by a V-shaped blunt tip, such that bottom portion **262C** forms a blunt bottom tip of epitaxial source/drain features **260**. In the X-Z plane, epitaxial source/drain features **260** have a top portion **264A** (a portion of epitaxial source/drain features **260** disposed above top surfaces of fins **222A**, **222B** along the z-direction) and a bottom portion **264B** (a portion of epitaxial source/drain features **260** disposed below top surfaces of fins **222A**, **222B** along the z-direction). In the depicted embodiment, top portion **264A** has a width w_{t1} that varies along its thickness t_t and bottom portion **264B** has a width w_{b1} that

varies along its thickness, which is substantially equal to depth d5 (i.e., target depth TD). In some embodiments, width wt1 of at least a portion of top portion **264A** is greater than a width of fins **222A**, **222B**. In some embodiments, top portion **264A** has a wide central portion with narrower end portions. In some embodiments, top portion **264A** has a substantially diamond-shaped profile. In some embodiments, top portion **264A** has a substantially hexagonal-shaped profile. In some embodiments, width wb1 of a first portion of bottom portion **264B** is substantially the same as a width of fins **222A**, **222B** and width b1 of a second portion of bottom portion **264B** decreases along the z-direction towards target depth TD.

[0048] Because of the three-step source/drain recess etch process described herein, epitaxial source/drain features **260** have tips T that are closer to top surfaces of fins **222A**, **222B** than can be achieved with conventional source/drain fabrication processes. For example, tips T have tip depth dt and tip proximity pt, advantages of which are described in detail herein. Epitaxial source/drain features **260** further have proximity footing pf that is greater than can be achieved with conventional source/drain fabrication processes, advantages of which are described in detail herein. In contrast to conventional source/drain fabrication processes, which define a location of tips after extending source/drain recesses to target depth TD, the proposed three-step source/drain recess etch process defines a location of tips T before extending source/drain recesses to target depth TD. Such fabrication technique raises the location of the tips T to locations that reduce parasitic resistance between epitaxial source/drain features **260** and subsequently formed metal gate while pulling back (in other words, increasing) proximity of a bottom portion of epitaxial source/drain features **260** to the subsequently formed metal gate (in particular, proximity footing pf), which reduces parasitic capacitance between epitaxial source/drain features **260** and the metal gate. Accordingly, epitaxial source/drain features **260** having the proposed candle light bulb-shaped profiles reduce DIBL, thereby improving short channel control of FinFET device **200**.

[0049] Further, the three-step source/drain etch process enables controlling (adjusting) a position (location) of tips T to about 10% of fin height FH to about 80% of fin height FH while pulling back bottom proximity of epitaxial source/drain features **260**. For example, as depicted in FIG. 5, whether depth dt of source/drain tips v1, v2 is about 10% of fin height FH (corresponding with source/drain recess **250-1** in FIG. 5) or about 80% of fin height FH (corresponding with source/drain recess **250-2** in FIG. 5), the three-step source/drain recess etch process produces proximity footing pf that is less than proximity footing pf at depths greater than or equal to about 80% of fin height FH, at which depths gate footing GF and corresponding capacitive coupling effects are most prevalent. Different embodiments may have different advantages, and no particular advantage is required of any embodiment.

[0050] Turning to FIG. 6, channel resistance and DIBL characteristics of FinFET devices having epitaxial source/drain features with various profiles are illustrated according to various aspects of the present disclosure. In FIG. 6, a graph **300** illustrates trade-offs between DIBL and channel resistance (Rch). For example, graph **300** correlates DIBL (in mV/V) (x-axis) with Rch (in ohms) (y-axis), where curve xx corresponds with FinFET devices having epitaxial source/drain features fabricated using conventional source/drain recess techniques and curve xy corresponds with FinFET devices having epitaxial source/drain features fabricated using the proposed three-step source/drain etch process. Both curve xx and curve xy indicate that as channel resistance decreases, for example, from pushing tip proximity closer to channel regions of the FinFET devices, DIBL increases, for example, from bottom portions of the epitaxial source/drain features pulling in closer to the channel regions of the FinFET devices as a result of the tip proximity push (thus leading to decreasing distances between the bottom portions of the epitaxial source/drain features and gate footing GF). In curve xx, R1 corresponds with channel resistance and DIBL characteristics of FinFET devices with epitaxial source/drain features having profiles achieved by an anisotropic etch that extends source/drain recesses to target depth TD (for example, substantially U-shaped profiles), and R2 corresponds with channel resistance and DIBL characteristics of FinFET devices

with epitaxial source/drain features having profiles achieved by an anisotropic etch that extends source/drain recesses to target depth followed by an isotropic etch to decrease tip proximity (and thus increase a width) of the source/drain recesses, such as those defined by conventional source/drain recesses **250'** depicted in FIG. 3H. As seen in FIG. 6, though conventional source/drain recesses **250'** achieve FinFET devices that exhibit less channel resistance, conventional source/drain recesses **250'** increase DIBL significantly because bottom portions of conventional source/drain recesses **250'** (in particular, the bottom tapered portions) are also pushed closer to the channel region during the isotropic etch. In curve xy, R3 corresponds with channel resistance and DIBL characteristics of FinFET devices with epitaxial source/drain features having profiles defined by the proposed three-step source/drain recess etch process, such as those defined by source/drain recesses **250**. As seen in FIG. 6, source/drain recesses **250** achieve FinFET devices having lower channel resistances comparable to those achieved by conventional source/drain recesses **250'**, but also recover DIBL increases suffered by FinFET devices with conventional source/drain recesses **250'**. FinFET devices having epitaxial source/drain features with profiles defined by source/drain recesses **250** thus exhibit reduced channel resistance and reduced DIBL compared to conventional FinFET devices, providing FinFET devices with better short channel control.

[0051] In some embodiments, as depicted in FIG. 7, epitaxial source/drain features **260** have a multi-layer structure, such as a first epitaxial semiconductor layer **260-1** disposed over a second epitaxial semiconductor layer **260-2**. First epitaxial semiconductor layer **260-1** and second epitaxial semiconductor layer **260-2** include at least one constituent that is the same, where a concentration of the constituent in first epitaxial semiconductor layer **260-1** is greater than a concentration of the constituent in second epitaxial semiconductor layer **260-2** to enhance conductivity and reduce resistance between epitaxial source/drain features **260** and subsequently formed contacts. For example, first epitaxial semiconductor layer **260-1** and second epitaxial semiconductor layer **260-2** each include silicon germanium, where a germanium concentration in first epitaxial semiconductor layer **260-1** is greater than a germanium concentration in second epitaxial semiconductor layer **260-2**. In some embodiments, a germanium concentration in first epitaxial semiconductor layer **260-1** is about 40% to about 60% and a germanium concentration in second epitaxial semiconductor layer **260-2** is about 20% to about 40%. In another example, first epitaxial semiconductor layer **260-1** and second epitaxial semiconductor layer **260-2** include the same material (for example, silicon or silicon germanium), where a dopant concentration of a dopant in first epitaxial semiconductor layer **260-1** (e.g., a dopant concentration of phosphorous or boron) is greater than a dopant concentration of the dopant in second epitaxial semiconductor layer **260-2**. In some embodiments, where first epitaxial semiconductor layer **260-1** and second epitaxial semiconductor layer **260-2** include silicon, a phosphorous dopant concentration in first epitaxial semiconductor layer **260-1** is about $1 \times 10^{21} \text{ cm}^{-3}$ to about $5 \times 10^{21} \text{ cm}^{-3}$ and a phosphorous dopant concentration in second epitaxial semiconductor layer **260-2** is about $1 \times 10^{20} \text{ cm}^{-3}$ to about $1 \times 10^{21} \text{ cm}^{-3}$. In some embodiments, where first epitaxial semiconductor layer **260-1** and second epitaxial semiconductor layer **260-2** include silicon germanium, a boron dopant concentration in first epitaxial semiconductor layer **260-1** is about $5 \times 10^{20} \text{ cm}^{-3}$ to about $1 \times 10^{21} \text{ cm}^{-3}$ and a boron dopant concentration in second epitaxial semiconductor layer **260-2** is about $1 \times 10^{20} \text{ cm}^{-3}$ to about $5 \times 10^{20} \text{ cm}^{-3}$. In some embodiments, second epitaxial semiconductor layer **260-2** is a buffer layer configured to reduce lattice mismatch between fins **222A**, **222B** and epitaxial source/drain features **260**. In such embodiments, a concentration of a constituent may be graded in second epitaxial semiconductor layer **260-2** from fins **222A**, **222B** to first epitaxial semiconductor layer **260-1**.

[0052] Turning to FIG. 3I and FIG. 4I, FinFET device **200** can undergo additional processing. For example, a dielectric layer **270** is formed over fins **222A**, **222B**, gate stacks **230A**, **230B**, gate spacers **242**, fin spacers **244**, and epitaxial source/drain features **260**. Dielectric layer **270** forms a

portion of a multi-layer interconnect (MLI) feature, in some embodiments. Dielectric layer **270** can include an interlayer dielectric (ILD) layer disposed over a contact etch stop layer (CESL). ILD layer includes a dielectric material including, for example, silicon oxide, carbon doped silicon oxide, silicon nitride, silicon oxynitride, tetraethyl orthosilicate (TEOS), PSG, BSG, boron-doped phosphosilicate glass (BPSG), fluorine-doped silicate glass (FSG), Black Diamond® (Applied Materials of Santa Clara, California), xerogel, aerogel, amorphous fluorinated carbon, parylene, benzocyclobutene (BCB)-based dielectric material, SILK (Dow Chemical, Midland, Michigan), polyimide, other suitable dielectric material, or combinations thereof. In some embodiments, ILD layer includes a low-k dielectric material, which generally refers to a dielectric material having a low dielectric constant relative to the dielectric constant of silicon dioxide ($k \approx 3.9$). For example, low-k dielectric material has a dielectric constant less than about 3.9. In some embodiments, the low-k dielectric material has a dielectric constant less than about 2.5, which can be referred to as an extreme low-k (ELK) dielectric material. In the depicted embodiment, ILD layer includes a low-k dielectric material. CESL includes a material different than ILD layer, such as a dielectric material that is different than the dielectric material of ILD layer. ILD layer and/or CESL can include a multilayer structure having multiple dielectric materials. In the depicted embodiment, where ILD layer includes silicon and oxygen (for example, SiCOH, SiO.sub.x, or other silicon-and-oxygen comprising material), CESL includes nitrogen and/or carbon (for example, SiN, SiCN, SiCON, SION, SiC, SiCO, metal nitride, and/or metal carbonitride). ILD layer and/or CESL are formed by a deposition process, such as CVD, PVD, ALD, HDPCVD, MOCVD, RPCVD, PECVD, LPCVD, ALCVD, APCVD, PEALD, other suitable methods, or combinations thereof. In some embodiments, ILD layer are formed by a flowable CVD (FCVD) process that includes, for example, depositing a flowable material (such as a liquid compound) over substrate **210** and converting the flowable material to a solid material by a suitable technique, such as thermal annealing and/or treating the flowable material with ultraviolet radiation. Subsequent to the deposition of ILD layer and/or CESL, a CMP process and/or other planarization process is performed, such that ILD layer and/or CESL have substantially planar surfaces.

[0053] In some embodiments, after forming dielectric layer **270**, gate stacks **230A**, **230B** are replaced with metal gate stacks **280A**, **280B** using any suitable gate replacement process, which includes various etching processes, deposition processes, lithography processes, and/or planarization processes. For example, gate stacks **230A**, **230B** are partially or fully removed (e.g., a dummy gate electrode layer, and in some embodiments, a dummy gate dielectric layer) is removed to form gate trenches (openings) that expose fin structures **222A**, **222B**, an interfacial layer, and/or gate dielectric layer and metal gate stacks **280A**, **280B** are formed in the gate trenches. Metal gate stacks **280A**, **280B** are configured to achieve desired functionality according to design requirements of FinFET device **200**, such that metal gate stack **280A** may include the same or different layers and/or materials as metal gate stack **280B**. In some embodiments, metal gate stacks **280A**, **280B** include a gate dielectric (for example, a gate dielectric layer) and a gate electrode (for example, a work function layer and a bulk conductive layer). Metal gate stacks **280A**, **280B** may include numerous other layers, for example, capping layers, interface layers, diffusion layers, barrier layers, hard mask layers, or combinations thereof. In some embodiments, the gate dielectric layer is disposed over an interfacial layer (including a dielectric material, such as silicon oxide), and the gate electrode is disposed over the gate dielectric layer. The gate dielectric layer includes a dielectric material, such as silicon oxide, high-k dielectric material, other suitable dielectric material, or combinations thereof. Examples of high-k dielectric material include hafnium dioxide (HfO.sub.2), HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, zirconium oxide, aluminum oxide, hafnium dioxide-alumina (HfO.sub.2—Al.sub.2O.sub.3) alloy, other suitable high-k dielectric materials, or combinations thereof. High-k dielectric material generally refers to dielectric materials having a high dielectric constant relative to a dielectric constant of silicon dioxide ($k \approx 3.9$). For example, high-k dielectric material has a dielectric constant greater than about 3.9. In some embodiments,

the gate dielectric layer is a high-k dielectric layer. The gate electrode includes a conductive material, such as polysilicon, Al, Cu, Ti, Ta, W, Mo, Co, TaN, NiSi, CoSi, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, other conductive material, or combinations thereof. In some embodiments, the work function layer is a conductive layer tuned to have a desired work function (e.g., an n-type work function or a p-type work function), and the conductive bulk layer is a conductive layer formed over the work function layer. In some embodiments, the work function layer includes n-type work function materials, such as Ti, Ag, Mn, Zr, TaAl, TaAlC, TiAlN, TaC, TaCN, TaSiN, other suitable n-type work function materials, or combinations thereof. In some embodiments, the work function layer includes a p-type work function material, such as Ru, Mo, Al, TiN, TaN, WN, ZrSi.sub.2, MoSi.sub.2, TaSi.sub.2, NiSi.sub.2, WN, other suitable p-type work function materials, or combinations thereof. The bulk (or fill) conductive layer includes a suitable conductive material, such as Al, W, and/or Cu. The bulk conductive layer may additionally or collectively include polysilicon, Ti, Ta, metal alloys, other suitable materials, or combinations thereof.

[0054] In some embodiments, various interconnects of the MLI feature are formed to facilitate operation of FinFET device **200**. MLI feature electrically couples various devices (for example, transistors, resistors, capacitors, and/or inductors) and/or components (for example, gate structures and/or source/drain features) of FinFET device **200**, such that the various devices and/or components can operate as specified by design requirements of FinFET device. MLI feature includes a combination of dielectric layers, such as dielectric layer **270**, and conductive layers configured to form various interconnects. During operation of FinFET device **200**, the interconnects are configured to route signals between the devices and/or the components of FinFET device **200** and/or distribute signals (for example, clock signals, voltage signals, and/or ground signals) to the devices and/or the components of FinFET device **200**. The conductive layers are configured to form vertical interconnects, such as device-level contacts and/or vias, and/or horizontal interconnects, such as conductive lines. Vertical interconnects typically connect horizontal interconnects in different layers (or different planes) of MLI feature. Device-level contacts (also referred to as local interconnects or local contacts) electrically couple and/or physically couple IC device features to other conductive features of the MLI feature, such as vias. Device-level contacts include metal-to-poly (MP) contacts, which generally refer to contacts to a gate structure, such as a poly gate structure or a metal gate structure, and metal-to-device (MD) contacts, which generally refer to contacts to a conductive region of FinFET device **200**, such as epitaxial source/drain features **260**. In FIG. 3I, an MD contact, such as source/drain contact **290**, extends through dielectric layer **270** to a respective epitaxial source/drain feature **260**. Source/drain contact **290** includes a conductive material, such as metal. Metals include aluminum, aluminum alloy (such as aluminum/silicon/copper alloy), copper, copper alloy, titanium, titanium nitride, tantalum, tantalum alloy, tantalum nitride, tungsten, tungsten alloy, cobalt, cobalt alloy, ruthenium, ruthenium alloy, polysilicon, metal silicide, other suitable metals, or combinations thereof. The metal silicide may include nickel silicide, cobalt silicide, tungsten silicide, tantalum silicide, titanium silicide, platinum silicide, erbium silicide, palladium silicide, or combinations thereof. In some implementations, a damascene process and/or dual damascene process is used to form a copper-based MLI feature.

[0055] In embodiments where epitaxial source/drain features **260** include a multi-layer structure, such as depicted in FIG. 7, the candle light bulb-shaped profile of epitaxial source/drain features **260** provides additional advantages. In some embodiments, source/drain contact **290** has a critical dimension D1 (e.g., a width along the y-direction or gate length direction) and epitaxial source/drain features **260** have a critical dimension D2 (e.g., a minimum width of first epitaxial semiconductor layer **260-1** in middle portion **262B** of epitaxial source/drain feature **260**). Because three-step source/drain etch process increases a width of source/drain recesses and defines a location of tips T before extending the source/drain recesses to target depth DT, source/drain recesses **250** are wider than conventional source/drain recesses **250'** at depths less than about 40%

of fin height FH. Critical dimension D2 of epitaxial source/drain features **260** is thus greater than critical dimension D3 (e.g., a minimum width of a first epitaxial semiconductor layer in a middle portion) of epitaxial source/drain features having profiles defined by conventional source/drain recesses **250'**, which reduces resistance between epitaxial source/drain features **260** and source/drain contact **290**. In some embodiments, critical dimension D2 is greater than critical dimension D1, whereas critical dimension D3 is substantially the same as critical dimension D1. In some embodiments, critical dimension D2 is greater than critical dimension D1 by at least 5 nm. In some embodiments, a distance between sidewalls of source/drain contact **290** and portions of first epitaxial semiconductor layer **260-1** defining critical dimension D2 is greater than or equal to about 2 nm. In some embodiments, in contrast to epitaxial source/drain features having profiles defined by conventional source/drain recesses **250'**, epitaxial first semiconductor layer **260-1** in middle portion **262B** of epitaxial source/drain features **260** does not extend beyond outer sidewalls OSs of gate spacers **242**.

[0056] The present disclosure provides for various source and drain formation techniques for fin-like field effect transistors (FinFETs). Source and drain formation techniques disclosed herein provide FinFETs with reduced channel resistance and reduced drain-induced barrier lowering. An exemplary three-step etch method for forming a source/drain recess in a source/drain region of a fin includes a first anisotropic etch, an isotropic etch, and a second anisotropic etch. The first anisotropic etch and the isotropic etch are tuned to define a location of a source/drain tip. A depth of the source/drain recess after the first anisotropic etch and the isotropic etch is less than a target depth. The second anisotropic etch is tuned to extend the depth of the source/drain recess to the target depth. The source/drain tip is near a top of the fin to reduce channel resistance while a bottom portion of the source/drain recess is spaced a distance from a gate footing that can minimize DIBL. The source/drain recess is filled with an epitaxial semiconductor material. In some embodiments, the first anisotropic etch process and the second anisotropic etch process implement a same etching gas. A first duration of the first anisotropic etch process may be different than a second duration of the second anisotropic etch process. In some embodiments, the first anisotropic etch process, the isotropic etch process, and the second anisotropic etch process are selective dry etch processes. In some embodiments, the first anisotropic etch process and the second anisotropic etch process implement a hydrogen-and-bromine-containing etch gas and the isotropic etch process implements a fluorine-and-chlorine-containing etch gas. In some embodiments, filling the source/drain recess with the epitaxial semiconductor material includes forming a first epitaxial semiconductor layer along surfaces of the fin defining the source/drain recess and forming a second epitaxial semiconductor layer over the first epitaxial semiconductor layer. The first epitaxial semiconductor layer has a first concentration of a constituent and the second epitaxial semiconductor layer has a second concentration of the constituent that is greater than the first concentration.

[0057] In some embodiment, the method further includes performing a third anisotropic etch process to form gate spacers. The third anisotropic etch process may begin formation of the source/drain recess. In some embodiments, the method includes tuning first parameters of the first anisotropic etch process and second parameters of the isotropic etch process to define the location of a source/drain tip and tuning third parameters of the second anisotropic etch process to extend the depth of the source/drain recess to the target depth. In some embodiments, the first parameters of the first anisotropic etch process are tuned to provide the source/drain recess with a U-shaped profile, the second parameters of the isotropic etch process are tuned to provide the source/drain recess with a bowl-shaped profile, and the third parameters of the second anisotropic etch process are tuned to provide the source/drain recess with a profile that resembles a light portion of a candle-shaped light bulb.

[0058] Another exemplary method includes forming a gate stack over a channel region of a fin and forming gate spacers along sidewalls of the gate stack. The channel region of the fin is disposed

between source/drain regions of the fin. The method further includes performing a first etching process to form source/drain recesses in the source/drain regions of the fin. The source/drain recesses have a first width and a first depth, and the first depth is less than a target depth for the source/drain recesses. The method further includes performing a second etching process to increase the first width of the source/drain recesses to a second width and performing a third etching process to increase the first depth of the source/drain recesses to the target depth. The method further includes forming semiconductor source/drain features in the source/drain recesses. In some embodiments, performing the first etching process and the third etching process includes removing fin material in substantially a vertical direction and performing the second etching process includes removing the fin material in substantially the vertical direction and substantially a horizontal direction. In some embodiments, the fin has a fin height, the first depth is equal to about 10% to about 80% of the fin height, and the target depth is equal to about 80% to about 120% of the fin height.

[0059] In some embodiments, performing the first etching process and the performing the second etching process includes controlling a tip depth and a tip proximity of the source/drain recesses along a fin height of the fin. In some embodiments, the fin has a fin height, each of the gate spacers has a gate spacer thickness, the first depth is less than about 50% of the fin height, the tip depth is about 10% of the fin height to about 30% of the fin height, and the tip proximity is about 60% of the gate spacer thickness to about 90% of the gate spacer thickness. In some embodiments, forming the gate spacers along the sidewalls of the gate stack includes forming a spacer layer over the gate stack and the source/drain regions of the fin, performing a first spacer etch, and performing a second spacer etch. The first spacer etch and the second spacer etch remove a portion of the spacer layer and begin formation of the source/drain recesses. The source/drain recesses have a second depth after the first spacer etch and a third depth after the second spacer etch. The second depth and the third depth are less than the first depth. In some embodiments, the fin has a fin height, the second depth is less than about 10% of the fin height, the third depth is about 10% to about 20% of the fin height, the first depth is about 10% to about 80% of the fin height, and the target depth is about 80% to about 120% of the fin height.

[0060] An exemplary device includes a fin having a fin height, a gate stack disposed over a channel region of the fin, a gate spacer disposed along a sidewall of the gate stack, and an epitaxial source/drain feature disposed in a source/drain region of the fin. The gate spacer has a gate spacer thickness. The epitaxial source/drain feature has a candle light bulb-shaped cross-sectional profile, such that the epitaxial source/drain feature includes a top portion disposed over a top surface of the fin, a middle portion disposed between the top surface of the fin and tips of the epitaxial source/drain feature, and a bottom portion disposed between the top surface of the fin and a depth of the epitaxial source/drain feature in the fin. A tip depth of the tips of the epitaxial source/drain feature is about 10% of the fin height to about 30% of the fin height. A proximity footing of the bottom portion of the epitaxial source/drain feature is greater than about 8 nm at depths greater than about 80% of the fin height. In some embodiments, a tip proximity of the tips of the epitaxial source/drain feature is about 60% of the gate spacer thickness to about 90% of the gate spacer thickness. In some embodiments, a first width of the top portion is substantially the same along a first thickness of the top portion, a second width of the middle portion increases along a second thickness of the middle portion, and a third width of the bottom portion decreases along a third thickness of the bottom portion. The first thickness, the second thickness, and the third thickness extend substantially parallel to a fin height direction. The third thickness may be greater than the second thickness and greater than the first thickness.

[0061] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages

of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device comprising: a channel structure disposed over a substrate, wherein a top surface of the channel structure is disposed at a height above the substrate along a first direction; a first source/drain structure and a second source/drain structure, wherein the channel structure extends lengthwise along a second direction from the first source/drain structure to the second source/drain structure and the second direction is different than the first direction; a gate stack disposed over and engaging the channel structure, wherein the gate stack is disposed between the first source/drain structure and the second source/drain structure along the second direction, and the gate stack has a gate footing; and wherein: a first sidewall of the first source/drain structure has a first tip disposed a first distance below the top surface of the channel structure, wherein the first distance is about 10% of the height to about 40% of the height, a second sidewall of the second source/drain structure has a second tip disposed a second distance below the top surface of the channel structure, wherein the second distance is about 10% of the height to about 40% of the height, a third distance between the first sidewall of the first source/drain structure and the gate footing is greater than about 8 nm at distances below the top surface of the channel structure that are greater than about 80% of the height, and a fourth distance between the second sidewall of the second source/drain structure and the gate footing is greater than about 8 nm at distances below the top surface of the channel structure that are greater than about 80% of the height.
2. The device of claim 1, wherein a minimum length of the channel structure along the second direction is between the first tip of the first source/drain structure and the second tip of the second source/drain structure.
3. The device of claim 2, wherein a length of the channel structure increases along the first direction from the minimum length to a maximum length, wherein the maximum length of the channel structure along the second direction is between a bottom of the first source/drain structure and a bottom of the second source/drain structure.
4. The device of claim 2, wherein the first distance is about the second distance.
5. The device of claim 1, wherein the first source/drain structure extends into the substrate and the second source/drain structure extends into the substrate, such that a bottommost tip of the first source/drain structure is disposed a fifth distance below the top surface of the channel structure and a bottommost tip of the second source/drain structure is disposed a sixth distance below the top surface of the channel structure, wherein each of the fifth distance and the sixth distance is greater than 100% of the height.
6. The device of claim 1, wherein the second source/drain structure includes: a first doped semiconductor layer; and a second doped semiconductor layer disposed over the first doped semiconductor layer, wherein a minimum width along the second direction of the second doped semiconductor layer above the second tip is greater than a width of a portion of a source/drain contact abutting the second source/drain structure.
7. The device of claim 6, wherein: the minimum width is at least 5 nm greater than the width of the portion of the source/drain contact abutting the second source/drain structure; and a fifth distance between the first doped semiconductor layer and sidewalls of the portion of the source/drain contact abutting the second source/drain structure is at least 2 nm.
8. The device of claim 6, wherein the first doped semiconductor layer has a first dopant concentration, the second doped semiconductor layer has a second dopant concentration, and the second dopant concentration is greater than the first dopant concentration.

9. The device of claim 1, wherein the gate stack has a gate height along the first direction, a gate length along the second direction, and a ratio of the gate height to the gate length is greater than about 5.

10. A device comprising: a channel structure disposed over a substrate; a first source/drain structure and a second source/drain structure, wherein the channel structure extends from the first source/drain structure to the second source/drain structure; a gate stack disposed over and engaging the channel structure, wherein the gate stack includes an upper portion having a first width and a lower portion having a second width, the second width is greater than the first width, and the lower portion is a gate footing; and wherein: a first proximity between the upper portion of the gate stack and the first source/drain structure at its maximum width is about 5 nm to about 8 nm, a second proximity between the lower portion of the gate stack and the first source/drain structure is greater than the first proximity, a third proximity between the upper portion of the gate stack and the second source/drain structure at its maximum width is about 5 nm to about 8 nm, a fourth proximity between the lower portion of the gate stack and the second source/drain structure is greater than the third proximity, and the second proximity is greater than about 6 nm and the fourth proximity is greater than about 6 nm.

11. The device of claim 10, wherein an aspect ratio of the gate stack is greater than about 5.

12. The device of claim 10, wherein: a first distance is between a top of the channel structure and a top of the substrate, wherein the first distance is along a gate height direction; and the first source/drain structure is at its maximum width and the second source/drain structure is at its maximum width at a second distance below the top of the channel structure, wherein the second distance is along the gate height direction and the second distance is about 10% of the first distance to about 40% of the first distance.

13. The device of claim 12, wherein the second distance below the top of the channel structure is about 10 nm to about 30 nm.

14. The device of claim 10, wherein: a first distance is between a top of the channel structure and a top of the substrate, wherein the first distance is along a gate height direction; the second proximity is greater than about 8 nm at distances below the top of the channel structure that are greater than about 80% of the first distance; and the fourth proximity is greater than about 8 nm at the distances below the top of the channel structure that are greater than about 80% of the first distance.

15. The device of claim 10, wherein the second width is tapered and the second width increases from the lower portion of the gate stack to a bottom of the gate stack.

16. The device of claim 15, wherein the first width is about the same from a top of the gate stack to the lower portion of the gate stack.

17. The device of claim 10, further comprising: a first gate spacer is disposed between the gate stack and the first source/drain structure, wherein the first gate spacer has a first thickness; a second gate spacer is disposed between the gate stack and the second source/drain structure, wherein the second gate spacer has a second thickness; and wherein the first proximity is about 50% to about 90% of the first thickness and the third proximity is about 50% to about 90% of the second thickness.

18. A method comprising: forming a source/drain trench by: performing a first etch that forms the source/drain trench with a first depth, wherein the first etch implements a first fluorine-containing etch gas and the first depth is along a gate height direction, performing a second etch to extend the source/drain trench to a second depth that is greater than the first depth, wherein the second etch implements a second fluorine-containing etch gas, performing a third etch to extend the source/drain trench to a third depth that is greater than the second depth, wherein the third etch implements a first hydrogen-and-bromine-containing etch gas, performing a fourth etch to enlarge the source/drain trench along a gate width direction, wherein the fourth etch implements a third fluorine-containing etch gas, and performing a fifth etch to extend the source/drain trench to a fourth depth that is greater than the third depth, wherein the fifth etch implements a second

hydrogen-and-bromine-containing etch gas and a first duration of the fifth etch is less than a second duration of the third etch; and forming source/drain semiconductor material in the source/drain trench.

19. The method of claim 18, wherein the first etch and the second etch remove portions of a dielectric layer to form gate spacers along sidewalls of a gate stack.

20. The method of claim 18, wherein: a first degree of anisotropy of the third etch is about 0.95 to about 1; a second degree of anisotropy of the fifth etch is about 0.95 to about 1; and a third degree of anisotropy of the fourth etch is about 0.8.
